

NCFM

NSE ACADEMY'S CERTIFICATION IN FINANCIAL MARKETS

OPTION TRADING STRATEGIES MODULE



NSE

ACADEMY

NCFM Module Examination Details

Sr. No	Module Name	Test Duration (in minutes)	No. Of Questions	Maximum Marks	Negative Marking	Pass marks	Allowable access to Candi- date at Test Centre			
							Open Office Spread Sheet	Nor- mal Distribution Table	Regular / Sci- entific Calcula- tor	Finan- cial Calcula- tor
FOUNDATION										
1	Financial Markets: A Beginners’ Module	120	60	100	NO	50	NO	NO	YES	NO
2	Mutual Funds : A Beginners' Module	120	60	100	NO	50	NO	NO	YES	NO
3	Currency Derivatives: A Beginner’s Module	120	60	100	NO	50	NO	NO	YES	NO
4	Equity Derivatives: A Beginner’s Module	120	60	100	NO	50	NO	NO	YES	NO
5	Interest Rate Derivatives: A Beginner's Module	120	60	100	NO	50	NO	NO	YES	NO
6	Commercial Banking in India: A Beginner's Module	120	60	100	NO	50	NO	NO	YES	NO
7	FIMMDA-NSE Debt Market (Basic) Module	120	60	100	YES	60	YES	NO	YES	NO
8	Securities Market (Basic) Module	120	60	100	YES	60	NO	NO	YES	NO
9	Clearing Settlement and Risk Management Module	60	75	100	NO	60	YES	NO	YES	NO
10	Banking Fundamental - International	90	48	48	YES	29	YES	NO	YES	NO
11	Capital Markets Fundamental - International	90	40	50	YES	30	YES	NO	YES	NO
INTERMEDIATE										
1	Capital Market (Dealers) Module	105	60	100	YES	50	NO	NO	YES	NO
2	Derivatives Market (Dealers) Module	120	60	100	YES	60	NO	NO	YES	NO
3	Investment Analysis and Portfolio Management	120	60	100	YES	60	NO	NO	YES	NO
4	Fundamental Analysis Module	120	60	100	YES	60	NO	NO	YES	NO
5	Operation Risk Management Module	120	75	100	YES	60	NO	NO	YES	NO
6	Options Trading Strategies Module	120	60	100	YES	60	NO	NO	YES	NO
7	Banking Sector Module	120	60	100	YES	60	NO	NO	YES	NO
8	Treasury Management Module	120	60	100	YES	60	YES	NO	YES	NO
9	Insurance Module	120	60	100	YES	60	NO	NO	YES	NO
10	Macroeconomics for Financial Markets Module	120	60	100	YES	60	NO	NO	YES	NO
11	NSDL-Depository Operations Module #	75	60	100	YES	60	NO	NO	YES	NO
12	Commodities Market Module	120	60	100	YES	50	NO	NO	YES	NO
13	Surveillance in Stock Exchanges Module	120	50	100	YES	60	NO	NO	YES	NO
14	Corporate Governance Module	90	100	100	YES	60	NO	NO	YES	NO
15	Compliance Officers (Brokers) Module	120	60	100	YES	60	NO	NO	YES	NO
16	Compliance Officers (Corporates) Module	120	60	100	YES	60	NO	NO	YES	NO
17	Information Security Auditors Module (Part-1)	120	90	100	YES	60	NO	NO	YES	NO
18	Information Security Auditors Module (Part-2)	120	90	100	YES	60	NO	NO	YES	NO
19	Technical Analysis Module	120	60	100	YES	60	NO	NO	YES	NO
20	Mergers and Acquisitions Module	120	60	100	YES	60	NO	NO	YES	NO
21	Back Office Operations Module	120	60	100	YES	60	NO	NO	YES	NO
22	Wealth Management Module	120	60	100	YES	60	NO	NO	YES	NO
23	Project Finance Module	120	60	100	YES	60	NO	NO	YES	NO
24	Venture Capital and Private Equity Module	120	70	100	YES	60	NO	NO	YES	NO
25	Financial Services Foundation Module ###	120	45	100	YES	50	NO	NO	YES	NO
26	NSE Certified Quality Analyst \$	120	60	100	YES	50	NO	NO	YES	NO
27	NSE's Capital Market Aptitude Test (NCMAT)	120	100	100	NO	60	YES	NO	YES	YES
29	NSE Certified Capital Market Professional (NCCMP)	120	60	100	NO	50	NO	NO	YES	NO
30	US Securities Operation Module	90	41	50	YES	30	YES	NO	YES	NO
ADVANCED										
1	Algorithmic Trading Module	120	100	100	YES	60	YES	NO	YES	NO
2	Financial Markets (Advanced) Module	120	60	100	YES	60	YES	NO	YES	NO
3	Securities Markets (Advanced) Module	120	60	100	YES	60	YES	NO	YES	NO
4	Derivatives (Advanced) Module	120	55	100	YES	60	YES	YES	YES	NO
5	Mutual Funds (Advanced) Module	120	60	100	YES	60	YES	NO	YES	NO
6	Options Trading (Advanced) Module	120	35	100	YES	60	YES	YES	YES	YES
7	Retirement Analysis and Investment Planning	120	77	150	NO	50	YES	NO	YES	YES
8	Retirement Planning and Employee Benefits **	120	77	150	NO	50	YES	NO	YES	YES
9	Tax Planning and Estate Planning **	120	77	150	NO	50	YES	NO	YES	YES
10	Investment Planning **	120	77	150	NO	50	YES	NO	YES	YES
11	Examination 5/Advanced Financial Planning **	240	30	100	NO	50	YES	NO	YES	YES
12	Equity Research Module ##	120	49	60	YES	60	YES	NO	YES	NO
13	Financial Valuation and Modeling	120	100	100	YES	60	YES	NO	YES	YES
14	Mutual Fund and Fixed Income Securities Module	120	100	60	YES	60	YES	NO	YES	YES
15	Issue Management Module ##	120	55	70	YES	60	YES	NO	YES	NO
16	Market Risk Module ##	120	40	65	YES	60	YES	NO	YES	NO
17	Financial Modeling Module ###	120	30	100	YES	50	YES	NO	YES	NO
18	Business Analytics Module ###	120	66	100	NO	50	YES	NO	YES	NO

Candidates securing 80% or more marks in NSDL-Depository Operations Module ONLY will be certified as 'Trainers'.

Module of IMS Preschool

Modules of Finitatives Learning India Pvt. Ltd. (FLIP

** Financial Planning Standards Board India (Certified Financial Planner Certification) FPSB India Exam

\$ SSA Business School

The curriculum for each of the modules (except Modules of Financial Planning Standards Board India, Finitatives Learning India Pvt. Ltd. and IMS Preschool) is available on our website: www.nseindia.com

Preface

About NSE Academy

NSE Academy is a subsidiary of National Stock Exchange of India. NSE Academy straddles the entire spectrum of financial courses for students of standard VIII and right up to MBA professionals. NSE Academy has tied up with premium educational institutes in order to develop pool of human resources having right skills and expertise which are apt for the financial market. Guided by our mission of spreading financial literacy for all, NSE Academy has constantly innovated its education template, this has resulted in improving the financial well-being of people at large in society. Our education courses have so far facilitated more than 41.8 lakh individuals become financially smarter through various initiatives.

NSE Academy Certification in Financial Markets (NCFM)

NCFM is an online certification programme aimed at upgrading skills and building competency.

The programme has a widespread reach with testing centres present at more than 154+ locations across the country.

The NCFM offers certifications ranging from the Basic to Advanced.

One can register for the NCFM through:

- Online mode by creating an online login id through the link 'Education' > 'Certifications' > 'Online Register / Enroll' available on the website www.nseindia.com
- Offline mode by filling up registration form available on the website www.nseindia.com > 'Education' > 'Certifications' > 'Register for Certification'

Once registered, a candidate is allotted a unique NCFM registration number along with an online login id and can avail of facilities like SMS alerts, online payment, checking of test schedules, online enrolment, profile update etc. through their login id.

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Distribution of weights of the Option Trading Strategies Curriculum

Chapter No.	Title	Weights (%)
1.	Introduction to Options	12
2.	Option Strategies	86
3.	Volatility	2

Note: Candidates are advised to refer to NSE's website: www.nseindia.com, click on 'Education' link and then go to 'Updates & Announcements' link, regarding revisions/updates in NCFM modules or launch of new modules, if any.

Only Open Office application will be made available on the PC for the tests of FIMMDA-NSE Debt Market (Basic) Module, NSE Academy Capital Market Aptitude Test (NCMAT), all the modules under the "Advanced Modules Category" and NISM Modules.

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National Stock Exchange of India Ltd. (NSE)

Exchange Plaza, Bandra Kurla Complex,

Bandra (East), Mumbai 400 051 INDIA

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OPTIONS

1. INTRODUCTION TO OPTIONS

An **option** is a contract written by a seller that conveys to the buyer the right — but not the obligation — to buy (in the case of a *call* option) or to sell (in the case of a *put* option) a particular asset, at a particular price (*Strike price / Exercise price*) in the future. In return for granting the option, the seller collects a payment (the *premium*) from the buyer. Exchange-traded options form an important class of options which have standardized contract features and trade on public exchanges, facilitating trading among a large number of investors. They provide settlement guarantee by the Clearing Corporation thereby reducing counterparty risk. Options can be used for hedging, taking a view on the future direction of the market, for arbitrage or for implementing strategies which can help in generating income for investors under various market conditions.

1.1. OPTION TERMINOLOGY

- **Index options:** These options have the index as the underlying. In India, they have a European style settlement. Eg. Nifty options, Mini Nifty options etc.
- **Stock options:** Stock options are options on individual stocks. A stock option contract gives the holder the right to buy or sell the underlying shares at the specified price. They have an American style settlement.
- **Buyer of an option:** The buyer of an option is the one who by paying the option premium buys the right but not the obligation to exercise his option on the seller/writer.
- **Writer/seller of an option:** The writer/seller of a call/put option is the one who receives the option premium and is thereby obliged to sell/buy the asset if the buyer exercises on him.
- **Call option:** A call option gives the holder the right but not the obligation to buy an asset by a certain date for a certain price.
- **Put option:** A put option gives the holder the right but not the obligation to sell an asset by a certain date for a certain price.
- **Option price/premium:** Option price is the price which the option buyer pays to the option seller. It is also referred to as the option premium.
- **Expiration date:** The date specified in the options contract is known as the expiration date, the exercise date, the strike date or the maturity.
- **Strike price:** The price specified in the options contract is known as the strike price or the exercise price.
- **American options:** American options are options that can be exercised at any time up to the expiration date.
- **European options:** European options are options that can be exercised only on the expiration date itself.

- ***In-the-money option:*** An in-the-money (ITM) option is an option that would lead to a positive cashflow to the holder if it were exercised immediately. A call option on the index is said to be in-the-money when the current index stands at a level higher than the strike price (i.e. spot price > strike price). If the index is much higher than the strike price, the call is said to be deep ITM. In the case of a put, the put is ITM if the index is below the strike price.
- ***At-the-money option:*** An at-the-money (ATM) option is an option that would lead to zero cashflow if it were exercised immediately. An option on the index is at-the-money when the current index equals the strike price (i.e. spot price = strike price).
- ***Out-of-the-money option:*** An out-of-the-money (OTM) option is an option that would lead to a negative cashflow if it were exercised immediately. A call option on the index is out-of-the-money when the current index stands at a level which is less than the strike price (i.e. spot price < strike price). If the index is much lower than the strike price, the call is said to be deep OTM. In the case of a put, the put is OTM if the index is above the strike price.
- ***Intrinsic value of an option:*** The option premium can be broken down into two components - intrinsic value and time value. The intrinsic value of a call is the amount the option is ITM, if it is ITM. If the call is OTM, its intrinsic value is zero. Putting it another way, the intrinsic value of a call is $\text{Max}[0, (S_t - K)]$ which means the intrinsic value of a call is the greater of 0 or $(S_t - K)$. Similarly, the intrinsic value of a put is $\text{Max}[0, K - S_t]$, i.e. the greater of 0 or $(K - S_t)$. K is the strike price and S_t is the spot price.
- ***Time value of an option:*** The time value of an option is the difference between its premium and its intrinsic value. Both calls and puts have time value. An option that is OTM or ATM has only time value. Usually, the maximum time value exists when the option is ATM. The longer the time to expiration, the greater is an option's time value, all else equal. At expiration, an option should have no time value.

1.2. OPTIONS PAYOFFS

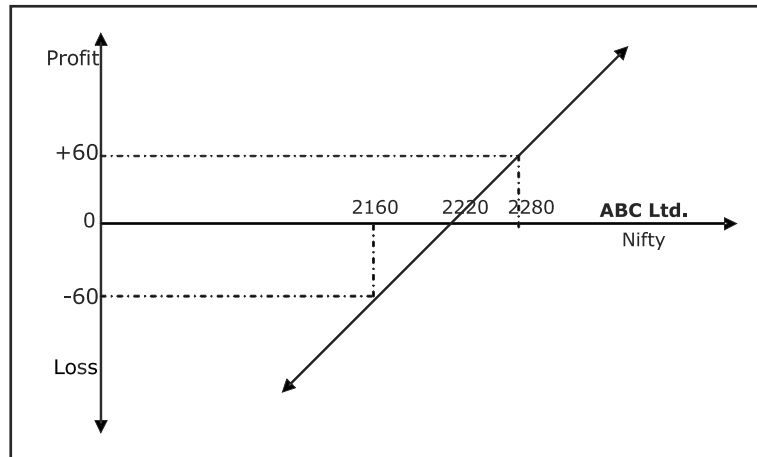
The optionality characteristic of options results in a non-linear payoff for options. In simple words, it means that the losses for the buyer of an option are limited; however, the profits are potentially unlimited. For a writer (seller), the payoff is exactly the opposite. His profits are limited to the option premium; however, his losses are potentially unlimited. These non-linear payoffs are fascinating as they lend themselves to be used to generate various payoffs by using combinations of options and the underlying. We look here at the six basic payoffs (pay close attention to these pay-offs, since all the strategies in the book are derived out of these basic payoffs).

1.2.1. Payoff profile of buyer of asset: Long asset

In this basic position, an investor buys the underlying asset, ABC Ltd. shares for instance, for ₹ 2220, and sells it at a future date at an unknown price, S_t . Once it is purchased, the investor is said to be "long" the asset. Figure 1.1 shows the payoff for a long position on ABC Ltd.

Figure 1.1 Payoff for investor who went Long ABC Ltd. at ₹ 2220

The figure shows the profits/losses from a long position on ABC Ltd. The investor bought ABC Ltd. at ₹ 2220. If the share price goes up, he profits. If the share price falls he loses.

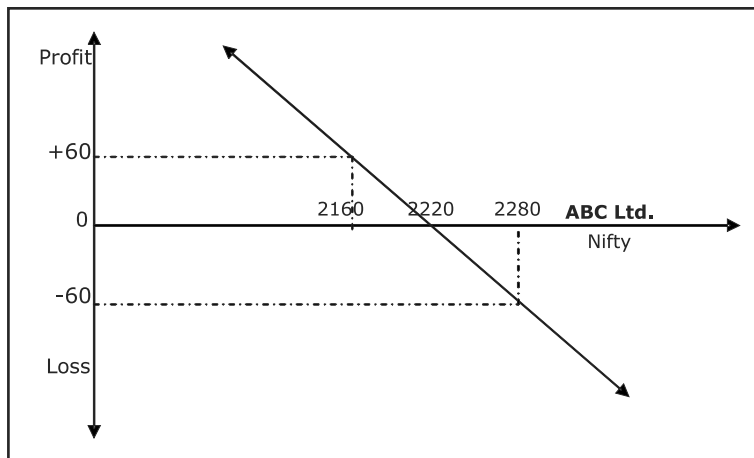


1.2.2. Payoff profile for seller of asset: Short asset

In this basic position, an investor shorts the underlying asset, ABC Ltd. shares for instance, for ₹ 2220, and buys it back at a future date at an unknown price, S_t . Once it is sold, the investor is said to be "short" the asset. Figure 1.2 shows the payoff for a short position on ABC Ltd...

Figure 1.2 Payoff for investor who went Short ABC Ltd. at ₹ 2220

The figure shows the profits/losses from a short position on ABC Ltd. The investor sold ABC Ltd. at ₹ 2220. If the share price falls, he profits. If the share price rises, he loses.

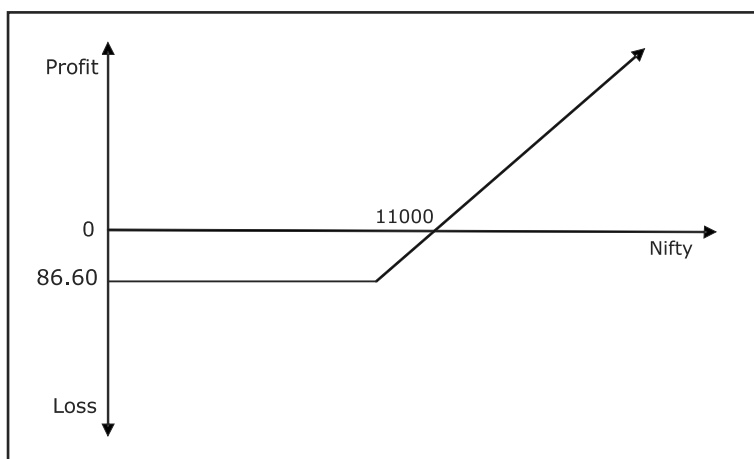


1.2.3. Payoff profile for buyer of call options: Long call

A call option gives the buyer the right to buy the underlying asset at the strike price specified in the option. The profit/loss that the buyer makes on the option depends on the spot price of the underlying. If upon expiration, the spot price exceeds the strike price, he makes a profit. Higher the spot price more is the profit he makes. If the spot price of the underlying is less than the strike price, he lets his option expire un-exercised. His loss, in this case, is the premium he paid for buying the option. Figure 1.3 gives the payoff for the buyer of a three-month call option (often referred to as long call) with a strike of 11000 bought at a premium of ₹ 86.60.

Figure 1.3 Payoff for buyer of call option

The figure shows the profits/losses for the buyer of a three-month Nifty 11000 call option. As can be seen, as the spot Nifty rises, the call option is in-the-money. If upon expiration, Nifty closes above the strike of 11000, the buyer would exercise his option and profit to the extent of the difference between the Nifty-close and the strike price. The profits possible with this option are potentially unlimited. However, if Nifty falls below the strike of 11000, he lets the option expire. His losses are limited to the extent of the premium he paid for buying the option.

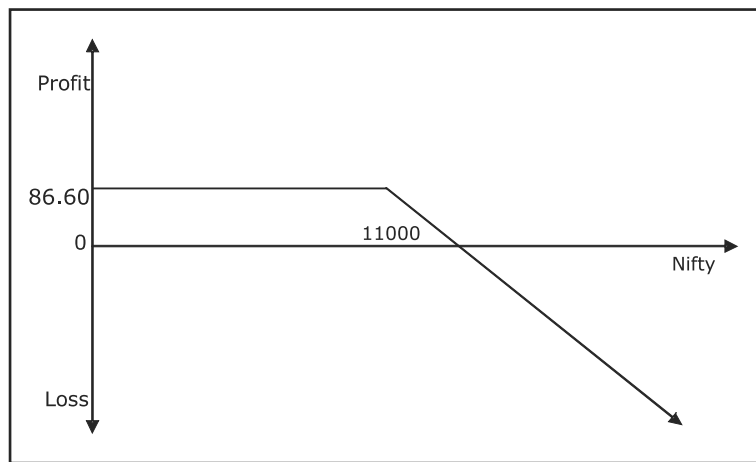


1.2.4. Payoff profile for writer (seller) of call options: Short call

A call option gives the buyer the right to buy the underlying asset at the strike price specified in the option. For selling the option, the writer of the option charges a premium. The profit/loss that the buyer makes on the option depends on the spot price of the underlying. Whatever is the buyer's profit is the seller's loss. If upon expiration, the spot price exceeds the strike price, the buyer will exercise the option on the writer. Hence as the spot price increases, the writer of the option starts making losses. Higher the spot price more is the loss he makes. If upon expiration the spot price of the underlying is less than the strike price, the buyer lets his option expire un-exercised and the writer gets to keep the premium. Figure 1.4 gives the payoff for the writer of a three-month call option (often referred to as short call) with a strike of 11000 sold at a premium of ₹ 86.60.

Figure 1.4 Payoff for buyer of call option

The figure shows the profits/losses for the seller of a three-month Nifty 11000 call option. As the spot Nifty rises, the call option is in-the-money and the writer starts making losses. If upon expiration, Nifty closes above the strike of 11000, the buyer would exercise his option on the writer who would suffer a loss to the extent of the difference between the Nifty-close and the strike price. The loss that can be incurred by the writer of the option is potentially unlimited, whereas the maximum profit is limited to the extent of the up-front option premium of ₹ 86.60 charged by him..

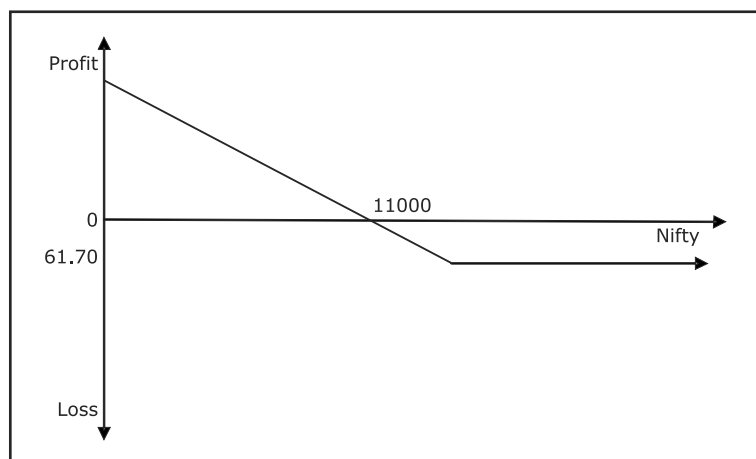


1.2.5. Payoff profile for buyer of put options: Long put

A put option gives the buyer the right to sell the underlying asset at the strike price specified in the option. The profit/loss that the buyer makes on the option depends on the spot price of the underlying. If upon expiration, the spot price is below the strike price, he makes a profit. Lower the spot price more is the profit he makes. If the spot price of the underlying is higher than the strike price, he lets his option expire un-exercised. His loss, in this case, is the premium he paid for buying the option. Figure 1.5 gives the payoff for the buyer of a three-month put option (often referred to as long put) with a strike of 11000 bought at a premium of ₹ 61.70.

Figure 1.5 Payoff for buyer of put option

The figure shows the profits/losses for the buyer of a three-month Nifty 11000 put option. As can be seen, as the spot Nifty falls, the put option is in-the-money. If upon expiration, Nifty closes below the strike of 11000, the buyer would exercise his option and profit to the extent of the difference between the strike price and Nifty-close. The profits possible on this option can be as high as the strike price. However if Nifty rises above the strike of 11000, he lets the option expire. His losses are limited to the extent of the premium he paid for buying the option.

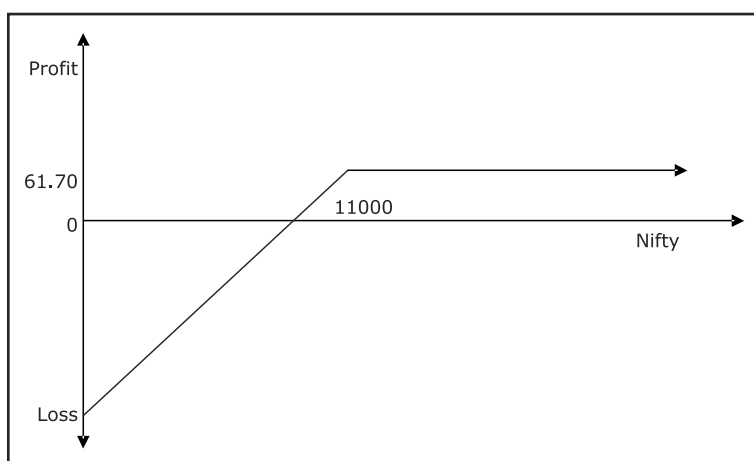


1.2.6 Payoff profile for writer (seller) of put options: Short put

A put option gives the buyer the right to sell the underlying asset at the strike price specified in the option. For selling the option, the writer of the option charges a premium. The profit/loss that the buyer makes on the option depends on the spot price of the underlying. Whatever is the buyer's profit is the seller's loss. If upon expiration, the spot price happens to be below the strike price, the buyer will exercise the option on the writer. If upon expiration the spot price of the underlying is more than the strike price, the buyer lets his option un-exercised and the writer gets to keep the premium. Figure 1.6 gives the payoff for the writer of a three-month put option (often referred to as short put) with a strike of 11000 sold at a premium of ₹ 61.70.

Figure 1.6 Payoff for writer of call option

The figure shows the profits/losses for the seller of a three-month Nifty 11000 put option. As the spot Nifty falls, the put option is in-the-money and the writer starts making losses. If upon expiration, Nifty closes below the strike of 11000, the buyer would exercise his option on the writer who would suffer a loss to the extent of the difference between the strike price and Nifty- close. The loss that can be incurred by the writer of the option is a maximum extent of the strike price (Since the worst that can happen is that the asset price can fall to zero) whereas the maximum profit is limited to the extent of the up-front option premium of ₹ 61.70 charged by him.



2. OPTIONS STRATEGIES

2.1. PROTECTIVE PUT

In this strategy, we purchase a stock since we feel bullish about it. However, to protect your investment if the stock price falls, you buy a Put Option on the stock. This gives you the right to sell the stock at a certain price which is the strike price of the Put Option. The strike price can be the price at which you bought the stock (ATM strike price) or lower (OTM strike price).

In case the price of the stock rises you get the full benefit of the price rise. However, if the price of the stock falls, exercise the Put Option (remember Put is a right to sell). You have capped your loss in this manner because the Put Option stops your further losses. It is a strategy with a limited loss and (after subtracting the Put premium) unlimited profit (from the stock price rise). The payoff of this strategy looks like a long Call Option and therefore, it is also called as Synthetic Call!

But the strategy is not to buy Call Option. Here you have taken an exposure to an underlying stock with the aim of holding it and reaping the benefits of price rise, dividends, bonus shares, rights issue, etc. and at the same time insuring against an adverse price movement. In simple buying of a Call Option, there is no underlying position in the stock but is entered into only to take advantage of price movement in the underlying stock.

When to use: When ownership is desired of stock yet investor is concerned about near-term downside risk. The outlook is **conservatively bullish**.

Risk: Losses limited to Stock price + Put Premium – Put Strike price

Reward: Profit potential is unlimited.

Break-even Point: Put Strike Price + Put Premium + Stock Price – Put Strike Price

Example

Mr. XYZ is bullish about ABC Ltd stock. He buys ABC Ltd. at current market price of ₹ 4000 on 4th July. To protect against fall in the price of ABC Ltd. (his risk), he buys an ABC Ltd. Put option with a strike price ₹ 3900 (OTM) at a premium of ₹ 143.80 expiring on 31st July.

Strategy : Buy Stock + Buy Put Option		
Buy Stock (Mr. XYZ pays)	Current Market Price of ABC Ltd. (₹)	4000
	Strike Price (₹)	3900
Buy Put (Mr. XYZ pays)	Premium (₹)	143.80
	Breakeven Point (₹) (Put Strike Price + Put Premium + Stock Price – Put Strike Price)*	4143.80

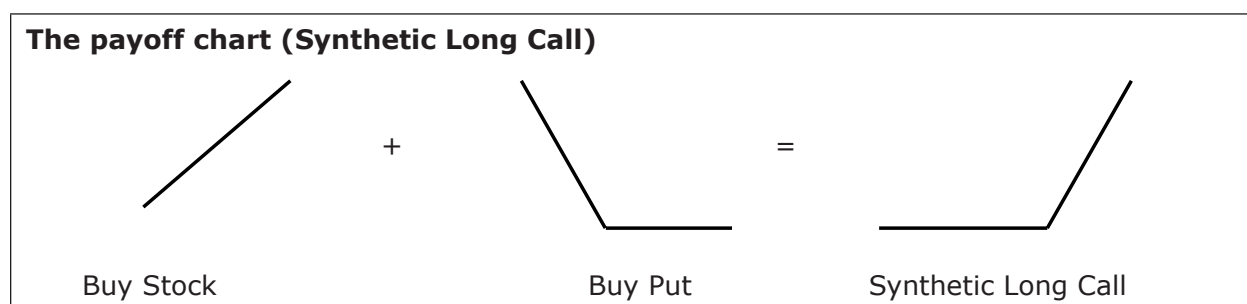
* Breakeven is from the point of view of Mr. XYZ. He has to recover the cost of the Put Option purchase price + the stock price to break even.

ABC Ltd. is trading at ₹ 4000 on 4th July. Buy 100 shares of the Stock at ₹ 4000. Buy 100 July Put Options with a Strike Price of ₹ 3900 at a premium of ₹ 143.80 per put.

Net Debit (payout)	Stock Bought + Premium Paid
	₹ 4000 + ₹ 143.80
	₹ 4,14,380/-
Maximum Loss	Stock Price + Put Premium – Put Strike
	₹ 4000 + ₹ 143.80 – ₹ 3900
	₹ 24,380
Maximum Gain	Unlimited (as the stock rises)
Breakeven	Put Strike + Put Premium + Stock Price – Put Strike
	₹ 3900 + ₹ 143.80 + ₹ 4000 – ₹ 3900
	= ₹ 4143.80

Payoff

ABC Ltd. closes at (₹) on expiry	Payoff from the Stock (₹)	Net Payoff from the Put Option (₹)	Net Payoff (₹)
3400	-600	356.2	-243.8
3600	-400	156.2	-243.8
3800	-200	-43.8	-243.8
4000	0	-143.8	-143.8
4143.8	143.8	-143.8	0
4200	200	-143.8	56.2
4400	400	-143.8	256.2
4600	600	-143.8	456.2
4800	800	-143.8	656.2



ANALYSIS: This is a low-risk strategy. This is a strategy which limits the loss in case of fall in the market but the potential profit remains unlimited when the stock price rises. A good strategy when you buy a stock for the medium or long term, with the aim of protecting any downside risk. The pay-off resembles a Call Option buy and is therefore called as Synthetic Long Call.

2.2. COVERED CALL

You own shares in a company which you feel may rise but not much in the near term (or at best stay sideways). You would still like to earn an income from the shares. The covered call is a strategy in which an investor Sells a Call option on a stock he owns (netting him a premium). The Call Option which is sold is usually an OTM Call. The Call would not get exercised unless the stock price increases above the strike price. Till then the investor in the stock (Call seller) can retain the Premium with him. This becomes his income from the stock. This strategy is usually adopted by a stock owner who is **Neutral to moderately Bullish** about the stock.

An investor buys a stock or owns a stock which he feels is good for medium to long term but is neutral or bearish for the near term. At the same time, the investor does not mind exiting the stock at a certain price (target price). The investor can sell a Call Option at the strike price at which he would be fine exiting the stock (OTM strike). By selling the Call Option the investor earns a Premium. Now the position of the investor is that of a Call Seller who owns the underlying stock. If the stock price stays at or below the strike price, the Call Buyer (refer to Strategy 1) will not exercise the Call. The Premium is retained by the investor.

In case the stock price goes above the strike price, the Call buyer who has the right to buy the stock at the strike price will exercise the Call option. The Call seller (the investor) who has to sell the stock to the Call buyer, will sell the stock at the strike price. This was the price which the Call seller (the investor) was any way interested in exiting the stock and now exits at that price. So besides the strike price which was the target price for selling the stock, the Call seller (investor) also earns the Premium which becomes an additional gain for him. This strategy is called as a Covered Call strategy because the Call sold is backed by a stock owned by the Call Seller (investor). The income increases as the stock rises, but gets capped after the stock reaches the strike price. Let us see an example to understand the Covered Call strategy.

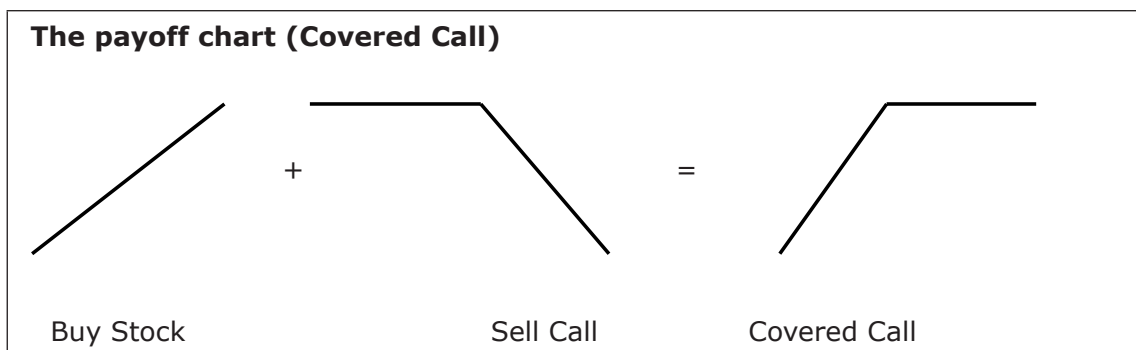
When to Use: This is often employed when an investor has a short-term neutral to moderately bullish view on the stock he holds. He takes a short position on the Call option to generate income from the option premium. Since the stock is purchased simultaneously with writing (selling) the Call, the strategy is commonly referred to as “buy-write”. Risk: If the Stock Price falls to zero, the investor loses the entire value of the Stock but retains the premium, since the Call will not be exercised against him. So maximum risk = Stock Price Paid – Call Premium Upside capped at the Strike price plus the Premium received. So if the Stock price rises beyond the Strike price the investor (Call seller) gives up all the gains on the stock. Reward: Limited to (Call Strike Price – Stock Price paid) + Premium received Breakeven: Stock Price paid - Premium Received	<u>Example</u> Mr. A bought XYZ Ltd. for ₹ 3850 and simultaneously sells a Call option at a strike price of ₹ 4000. Which means Mr. A does not think that the price of XYZ Ltd. will rise above ₹ 4000. However, in case it rises above ₹ 4000, Mr. A does not mind getting exercised at that price and exiting the stock at ₹ 4000 (TARGET SELL PRICE = 3.90% return on the stock purchase price). Mr A receives a premium of ₹ 80 for selling the Call. Thus net outflow to Mr. A is (₹ 3850 – ₹ 80) = ₹ 3770. He reduces the cost of buying the stock by this strategy. If the stock price stays at or below ₹ 4000, the Call option will not get exercised and Mr. A can retain the ₹ 80 premium, which is an extra income. If the stock price goes above ₹ 4000, the Call option will get exercised by the Call buyer. The entire position will work like this: <table><tr><th colspan="3">Strategy : Buy Stock + Sell Call Option</th></tr><tr><td>Mr. A buys the stock XYZ Ltd.</td><td>Market Price (₹)</td><td>3850</td></tr><tr><td>Call Options</td><td>Strike Price (₹)</td><td>4000</td></tr><tr><td>Mr. A receives</td><td>Premium (₹)</td><td>80</td></tr><tr><td></td><td>Breakeven Point (₹) (Stock Price paid - Premium Received)</td><td>3770</td></tr></table>	Strategy : Buy Stock + Sell Call Option			Mr. A buys the stock XYZ Ltd.	Market Price (₹)	3850	Call Options	Strike Price (₹)	4000	Mr. A receives	Premium (₹)	80		Breakeven Point (₹) (Stock Price paid - Premium Received)	3770
Strategy : Buy Stock + Sell Call Option																
Mr. A buys the stock XYZ Ltd.	Market Price (₹)	3850														
Call Options	Strike Price (₹)	4000														
Mr. A receives	Premium (₹)	80														
	Breakeven Point (₹) (Stock Price paid - Premium Received)	3770														

- 1) The price of XYZ Ltd. stays at or below ₹ 4000. The Call buyer will not exercise the Call Option. Mr. A will keep the premium of ₹ 80. This is an income for him. So if the stock has moved from ₹ 3850 (purchase price) to ₹ 3950, Mr. A makes ₹ 180/- [₹ 3950 – ₹ 3850 + ₹ 80 (Premium)] = an additional ₹ 80, because of the Call sold.
- 2) Suppose the price of XYZ Ltd. moves to ₹ 4100, then the Call Buyer will exercise the Call Option and Mr. A will have to pay him ₹ 100 (loss on exercise of the Call Option). What would Mr. A do and what will be his pay – off?
 - a) Sell the Stock in the market at : ₹ 4100
 - b) Pay ₹ 100 to the Call Options buyer : - ₹ 100
 - c) Pay Off (a – b) received : ₹ 4000
(This was Mr. A's target price)
 - d) Premium received on Selling Call Option : ₹ 80

e) Net payment (c + d) received by Mr. A	:	₹ 4080
f) Purchase price of XYZ Ltd.	:	₹ 3850
g) Net profit	:	₹ 4080 – ₹ 3850 = ₹ 230
h) Return (%)	:	$\frac{(\text{₹ } 4080 - \text{₹ } 3850) \times 100}{\text{₹ } 3850}$ = 5.97% (which is more than the target return of 3.90%).

Payoff

XYZ Ltd. price closes at (₹)	Net Payoff (₹)
3600	-170
3700	-70
3740	-30
3770	0
3800	30
3900	130
4000	230
4100	230
4200	230
4300	230



2.3. LONG COMBO (SELL A PUT AND BUY A CALL)

A Long Combo is a **Bullish** strategy. If an investor is expecting the price of a stock to move up he can do a Long Combo strategy. It involves selling an OTM (lower strike) Put and buying an OTM (higher strike) Call. This strategy simulates the action of buying a stock (or futures) but at a fraction of the stock price. It is an inexpensive trade, similar in pay-off to Long Stock, except there is a gap between the strikes (please see the payoff diagram). As the stock price rises the strategy starts making profits. Let us try and understand Long Combo with an example.

When to Use: Investor is **bullish** on the stock.

Risk: Unlimited (Lower Strike + net debit)

Reward: Unlimited

Breakeven:

Higher strike + net debit

Example:

A stock ABC Ltd. is trading at ₹ 450. Mr. XYZ is bullish on the stock. But does not want to invest ₹ 450. He does a Long Combo. He sells a Put option with a strike price ₹ 400 at a premium of ₹ 1.00 and buys a Call Option with a strike price of ₹ 500 at a premium of ₹ 2. The net cost of the strategy (net debit) is ₹ 1.

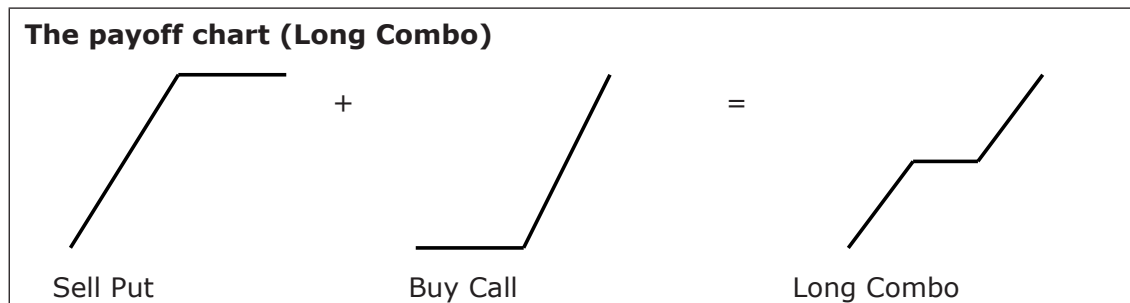
Strategy: Sell a Put + Buy a Call		
ABC Ltd.	Current Market Price (₹)	450
Sells Put	Strike Price (₹)	400
Mr. XYZ receives	Premium (₹)	1.00
Buys Call	Strike Price (₹)	500
Mr. XYZ pays	Premium (₹)	2.00
	Net Debit (₹)	1.00
	Breakeven Point (₹) (Higher Strike + Net Debit)	₹ 501.00

Payoff

ABC Ltd. closes at (₹)	Net Payoff from the Put Sold (₹)	Net Payoff from the Call purchased (₹)	Net Payoff (₹)
700	1	198	199
650	1	148	149
600	1	98	99
550	1	48	49
501	1	-1	0
500	1	-2	-1
450	1	-2	-1
400	1	-2	-1
350	-49	-2	-51
300	-99	-2	-101
250	-149	-2	-151

For a small investment of ₹ 1 (net debit), the returns can be very high in a Long Combo, but only if the stock moves up. Otherwise the potential losses can also be high.

The payoff chart (Long Combo)



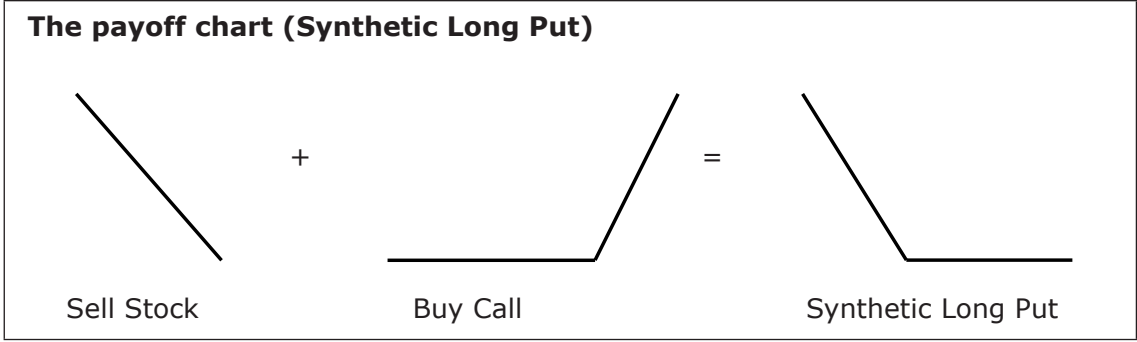
2.4. PROTECTIVE CALL/SYNTHETIC LONG PUT

This is a strategy wherein an investor has gone short on a stock and buys a call to hedge. This is an opposite of Synthetic Call (Strategy 3). An investor shorts a stock and buys an ATM or slightly OTM Call. The net effect of this is that the investor creates a pay-off like a Long Put, but instead of having a net debit (paying premium) for a Long Put, he creates a net credit (receives money on shorting the stock). In case the stock price falls the investor gains in the downward fall in the price. However, in case there is an unexpected rise in the price of the stock the loss is limited. The pay-off from the Long Call will increase thereby compensating for the loss in value of the short stock position. This strategy hedges the upside in the stock position while retaining downside profit potential.

When to Use: If the investor is of the view that the markets will go down (bearish) but wants to protect against any unexpected rise in the price of the stock. Risk: Limited. Maximum Risk is Call Strike Price – Stock Price + Premium Reward: Maximum is Stock Price – Call Premium Breakeven: Stock Price – Call Premium	Example: Suppose ABC Ltd. is trading at ₹ 4457 in June. An investor Mr. A buys a ₹ 4500 call for ₹ 100 while shorting the stock at ₹ 4457. The net credit to the investor is ₹ 4357 (₹ 4457 – ₹ 100). <table><tr><th colspan="3">Strategy : Short Stock + Buy Call Option</th></tr><tr><td>Sells Stock (Mr. A receives)</td><td>Current Market Price (₹)</td><td>4457</td></tr><tr><td>Buys Call</td><td>Strike Price (₹)</td><td>4500</td></tr><tr><td>Mr. A pays</td><td>Premium (₹)</td><td>100</td></tr><tr><td></td><td>Break Even Point (₹) (Stock Price – Call Premium)</td><td>4357</td></tr></table>	Strategy : Short Stock + Buy Call Option			Sells Stock (Mr. A receives)	Current Market Price (₹)	4457	Buys Call	Strike Price (₹)	4500	Mr. A pays	Premium (₹)	100		Break Even Point (₹) (Stock Price – Call Premium)	4357
Strategy : Short Stock + Buy Call Option																
Sells Stock (Mr. A receives)	Current Market Price (₹)	4457														
Buys Call	Strike Price (₹)	4500														
Mr. A pays	Premium (₹)	100														
	Break Even Point (₹) (Stock Price – Call Premium)	4357														

Payoff

ABC Ltd. closes at (₹)	Payoff from the stock (₹)	Net Payoff from the Call Option (₹)	Net Payoff (₹)
4100	357	-100	257
4150	307	-100	207
4200	257	-100	157
4300	157	-100	57
4350	107	-100	7
4357	100	-100	0
4400	57	-100	-43
4457	0	-100	-100
4600	-143	0	-143
4700	-243	100	-143
4800	-343	200	-143
4900	-443	300	-143
5000	-543	400	-143



2.5. COVERED PUT

This strategy is opposite to a Covered Call. A Covered Call is a neutral to bullish strategy, whereas a Covered Put is a neutral to **Bearish** strategy. You do this strategy when you feel the price of a stock/index is going to remain range bound or move down. Covered Put writing involves a short in a stock/index along with a short Put on the options on the stock /index.

The Put that is sold is generally an OTM Put. The investor shorts a stock because he is bearish about it, but does not mind buying it back once the price reaches (falls to) a target price. This target price is the price at which the investor shorts the Put (Put strike price). Selling a Put means, buying the stock at the strike price if exercised (Strategy no. 2). If the stock falls below the Put strike, the option will be exercised and the investor will have to buy the stock at the strike price (which is anyway his target price to repurchase the stock). The investor makes a profit because he has shorted the stock and purchasing it at the strike price simply closes the short stock position at a profit. And the investor keeps the Premium on the Put sold. The investor is covered here because he shorted the stock in the first place.

If the stock price does not change, the investor gets to keep the Premium. He can use this strategy as an income in a neutral market. Let us understand this with an example.

When to Use: If the investor is of the view that the markets are **moderately bearish**.

Risk: Unlimited if the price of the stock rises substantially

Reward: Maximum is (Sale Price of the Stock – Strike Price) + Put Premium

Breakeven: Sale Price of Stock + Put Premium

Example:

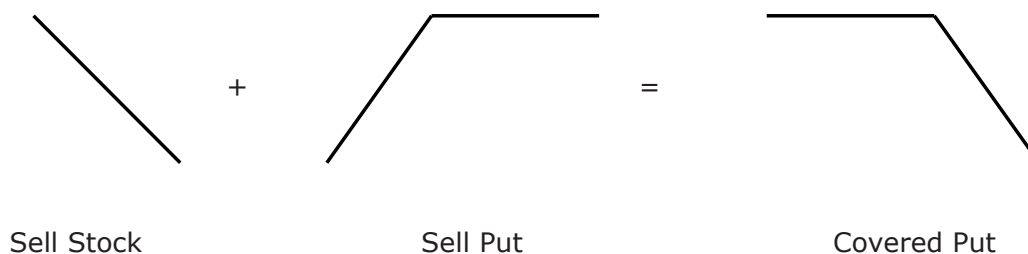
Suppose ABC Ltd. is trading at ₹ 4500 in June. An investor, Mr. A, shorts ₹ 4300 Put by selling a July Put for ₹ 24 while shorting an ABC Ltd. stock. The net credit received by Mr. A is ₹ 4500 + ₹ 24 = ₹ 4524.

Strategy : Short Stock + Short Put Option		
Sells Stock (Mr. A receives)	Current Market Price (₹)	4500
Sells Put	Strike Price (₹)	4300
Mr. A receives	Premium (₹)	24
	Break Even Point (₹) (Sale price of Stock + Put Premium)	4524

Payoff

ABC Ltd. closes at (₹)	Payoff from the stock (₹)	Net Payoff from the Put Option (₹)	Net Payoff (₹)
4000	500	-276	224
4100	400	-176	224
4200	300	-76	224
4300	200	24	224
4400	100	24	124
4450	50	24	74
4500	0	24	24
4524	-24	24	0
4550	-50	24	-26
4600	-100	24	-76
4635	-135	24	-111
4650	-160	24	-136

The payoff chart (Covered Put)



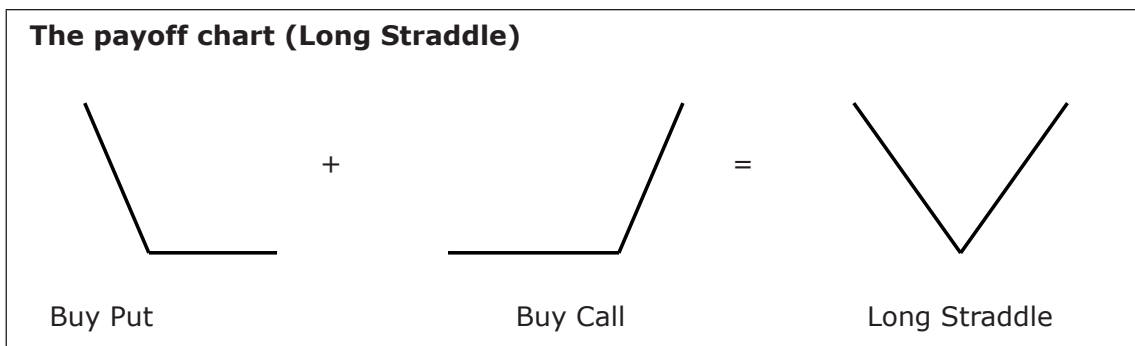
2.6. LONG STRADDLE

A Straddle is a volatility strategy and is used when the stock price/index is expected to show large movements. This strategy involves buying a call as well as put on the same stock/index for the same maturity and strike price, to take advantage of a movement in either direction, a soaring or plummeting value of the stock/index. If the price of the stock /index increases, the call is exercised while the put expires worthless and if the price of the stock/index decreases, the put is exercised, the call expires worthless. Either way, if the stock/index shows volatility to cover the cost of the trade, profits are to be made. With Straddles, the investor is direction neutral. All that he is looking out for is the stock/index to break out exponentially in either direction.

When to Use: The investor thinks that the underlying stock/index will experience significant volatility in the near term. Risk: Limited to the initial premium paid. Reward: Unlimited Breakeven: - Upper Breakeven Point = Strike Price of Long Call + Net Premium Paid - Lower Breakeven Point = Strike Price of Long Put – Net Premium Paid	Example Suppose Nifty is at 11000 on 31st January. An investor, Mr. A enters a long straddle by buying a February ₹ 11000 Nifty Put for ₹ 122 and a February ₹ 11000 Nifty Call for ₹ 85. The net debit taken to enter the trade is ₹ 207, which is also his maximum possible loss. <table><tr><th colspan="3">Strategy: Buy Put + Buy Call</th></tr><tr><td>Nifty index</td><td>Current Value</td><td>11000</td></tr><tr><td>Call and Put</td><td>Strike Price (₹)</td><td>11000</td></tr><tr><td>Mr. A pays</td><td>Total Premium (Call + Put) (₹)</td><td>207</td></tr><tr><td></td><td>Breakeven Point Upper (₹)</td><td>11207</td></tr><tr><td></td><td>Lower (₹)</td><td>10793</td></tr></table>	Strategy: Buy Put + Buy Call			Nifty index	Current Value	11000	Call and Put	Strike Price (₹)	11000	Mr. A pays	Total Premium (Call + Put) (₹)	207		Breakeven Point Upper (₹)	11207		Lower (₹)	10793
Strategy: Buy Put + Buy Call																			
Nifty index	Current Value	11000																	
Call and Put	Strike Price (₹)	11000																	
Mr. A pays	Total Premium (Call + Put) (₹)	207																	
	Breakeven Point Upper (₹)	11207																	
	Lower (₹)	10793																	

Payoff

On expiry Nifty closes at	Net Payoff from Put purchased (₹)	Net Payoff from Call purchased (₹)	Net Payoff (₹)
10100	778	-85	693
10200	678	-85	593
10300	578	-85	493
10400	478	-85	393
10500	378	-85	293
10600	278	-85	193
10700	178	-85	93
10793	85	-85	0
10900	-22	-85	-107
11000	-122	-85	-207
11100	-122	15	-107
11207	-122	122	0
11300	-122	215	93
11400	-122	315	193
11500	-122	415	293
11600	-122	515	393
11700	-122	615	493
11800	-122	715	593
11900	-122	815	693
12000	-122	915	793



2.7. SHORT STRADDLE

A Short Straddle is the opposite of Long Straddle. It is a strategy to be adopted when the investor feels the market will not show much movement. He sells a Call and a Put on the same stock/index for the same maturity and strike price. It creates a net income for the investor. If the stock/index does not move much in either direction, the investor retains the Premium as neither the Call nor the Put will be exercised. However, in case the stock/index moves in either direction, up or down significantly, the investor's losses can be significant. So this is a risky strategy and should be carefully adopted and only when the expected volatility in the market is limited. If the stock/index value stays close to the strike price on expiry of the contracts, maximum gain, which is the Premium received is made.

When to Use: The investor thinks that the underlying stock/index will experience **very little volatility** in the near term.

Risk: Unlimited

Reward: Limited to the premium received

Breakeven:

- Upper Breakeven Point = Strike Price of Short Call + Net Premium Received
- Lower Breakeven Point = Strike Price of Short Put - Net Premium Received

Example

Suppose Nifty is at 11000 on 31st January. An investor, Mr. A, enters into a short straddle by selling a February ₹ 11000 Nifty Put for ₹ 122 and a February ₹ 11000 Nifty Call for ₹ 85. The net credit received is ₹ 207, which is also his maximum possible profit.

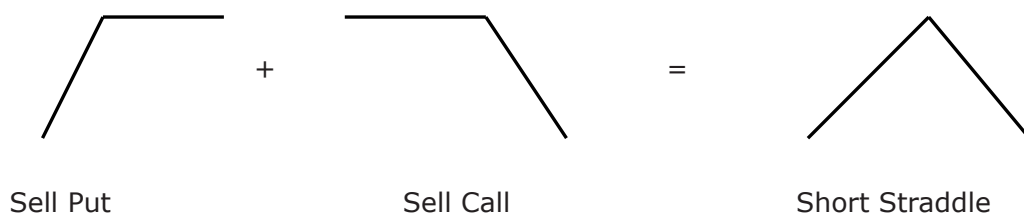
Strategy : Sell Put + Sell Call		
Nifty index	Current Value	11000
Call and Put	Strike Price (₹)	11000
Mr. A receives	Total Premium (Call + Put) (₹)	207
	Breakeven Point (₹)*	11207(U)
	(₹)*	10793(L)

* From buyer's point of view

Payoff

On expiry Nifty closes at	Net Payoff from Put sold (₹)	Net Payoff from Call sold (₹)	Net Payoff (₹)
10100	-778	85	-693
10200	-678	85	-593
10300	-578	85	-493
10400	-478	85	-393
10500	-378	85	-293
10600	-278	85	-193
10700	-178	85	-93
10793	-85	85	0
10900	22	85	107
11000	122	85	207
11100	122	-15	107
11207	122	-122	0
11300	122	-215	-93
11400	122	-315	-193
11500	122	-415	-293
11600	122	-515	-393
11700	122	-615	-493
11800	122	-715	-593
11900	122	-815	-693
12000	122	-915	-793

The payoff chart (Short Straddle)



Payoff

On expiry Nifty closes at	Net Payoff from Put purchased (₹)	Net Payoff from Call purchased (₹)	Net Payoff (₹)
10100	657	-23	634
10200	557	-23	534
10300	457	-23	434
10400	357	-23	334
10500	257	-23	234
10600	157	-23	134
10700	57	-23	34
10757	0	-23	-23
10900	-43	-23	-66
11000	-43	-23	-66
11100	-43	-23	-66
11223	-43	0	-43
11300	-43	77	34
11400	-43	177	134
11500	-43	277	234
11600	-43	377	334
11700	-43	477	434
11800	-43	577	534
11900	-43	677	634
12000	-43	777	734

The payoff chart (Long Strangle)



2.9. SHORT STRANGLE

A Short Strangle is a slight modification to the Short Straddle. It tries to improve the profitability of the trade for the Seller of the options by widening the breakeven points so that there is a much greater movement required in the underlying stock/index, for the Call and Put option to be worth exercising. This strategy involves the simultaneous selling of a slightly out-of-the-

money (OTM) put and a slightly out-of-the-money (OTM) call of the same underlying stock and expiration date. This typically means that since OTM call and put are sold, the net credit received by the seller is less as compared to a Short Straddle, but the break-even points are also widened. The underlying stock has to move significantly for the Call and the Put to be worth exercising. If the underlying stock does not show much of a movement, the seller of the Strangle gets to keep the Premium.

When to Use: This options trading strategy is taken when the options investor thinks that the underlying stock will experience little **volatility** in the near term.

Risk: Unlimited

Reward: Limited to the premium received

Breakeven:

- Upper Breakeven Point = Strike Price of Short Call + Net Premium Received
- Lower Breakeven Point = Strike Price of Short Put - Net Premium Received

Example

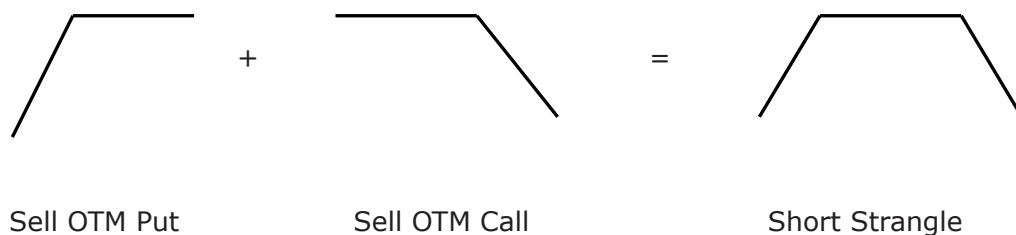
Suppose Nifty is at 11000 in January. An investor, Mr. A, executes a Short Strangle by selling a ₹ 10800 Nifty Put for a premium of ₹ 43 and a ₹ 11200 Nifty Call for ₹ 23. The net credit is ₹ 66, which is also his maximum possible gain.

Strategy : Sell OTM Put + Sell OTM Call		
Nifty index	Current Value	11000
Sell Call Option	Strike Price (₹)	11200
Mr. A receives	Premium (₹)	23
	Break Even Point (₹)	11223
Sell Put Option	Strike Price (₹)	10800
Mr. A receives	Premium (₹)	43
	Breakeven Point (₹)	10757

Payoff

On expiry Nifty closes at	Net Payoff from Put purchased (₹)	Net Payoff from Call purchased (₹)	Net Payoff (₹)
10100	-657	23	-634
10200	-557	23	-534
10300	-457	23	-434
10400	-357	23	-334
10500	-257	23	-234
10600	-157	23	-134
10700	-57	23	-34
10793	0	23	23
10900	43	23	66
11000	43	23	66
11100	43	23	66
11207	43	0	43
11300	43	-77	-34
11400	43	-177	-134
11500	43	-277	-234
11600	43	-377	-334
11700	43	-477	-434
11800	43	-577	-534
11900	43	-677	-634
12000	43	-777	-734

The payoff chart (Short Strangle)



2.10. COLLAR

A Collar is similar to Covered Call but involves another leg – buying a Put to insure against the fall in the price of the stock. It is a Covered Call with a limited risk. So a Collar is buying a stock, insuring against the downside by buying a Put and then financing (partly) the Put by selling a Call.

The put generally is ATM and the call is OTM having the same expiration month and must be

equal in the number of shares. This is a low-risk strategy since the Put prevents downside risk. However, do not expect unlimited rewards since the Call prevents that. It is a strategy to be adopted when the investor is **conservatively bullish**. The following example should make Collar easier to understand.

When to Use: The collar is a good strategy to use if the investor is writing covered calls to earn premiums but wishes to protect him from an unexpectedly sharp drop in the price of the underlying security. Risk: Limited Reward: Limited Breakeven: Purchase Price of Underlying – Call Premium + Put Premium	Example Suppose an investor Mr. A buys or is holding ABC Ltd. currently trading at ₹ 4758. He decides to establish a collar by writing a Call of strike price ₹ 5000 for ₹ 39 while simultaneously purchasing a ₹ 4700 strike price Put for ₹ 27. Since he pays ₹ 4758 for the stock ABC Ltd., another ₹ 27 for the Put but receives ₹ 39 for selling the Call option, his total investment is ₹ 4746. <table><tr><th colspan="3">Strategy : Buy Stock + Buy Put + Sell Call</th></tr><tr><td>ABC Ltd.</td><td>Current Market Price (₹)</td><td>4758</td></tr><tr><td>Sell Call Option</td><td>Strike Price (₹)</td><td>5000</td></tr><tr><td>Mr. A Receives</td><td>Premium (₹)</td><td>39</td></tr><tr><td>Buy Put Option</td><td>Strike Price (₹)</td><td>4700</td></tr><tr><td>Mr. A Pays</td><td>Premium (₹)</td><td>27</td></tr><tr><td></td><td>Net Premium Received (₹)</td><td>12</td></tr><tr><td></td><td>Breakeven Point (₹)</td><td>4746</td></tr></table>	Strategy : Buy Stock + Buy Put + Sell Call			ABC Ltd.	Current Market Price (₹)	4758	Sell Call Option	Strike Price (₹)	5000	Mr. A Receives	Premium (₹)	39	Buy Put Option	Strike Price (₹)	4700	Mr. A Pays	Premium (₹)	27		Net Premium Received (₹)	12		Breakeven Point (₹)	4746
Strategy : Buy Stock + Buy Put + Sell Call																									
ABC Ltd.	Current Market Price (₹)	4758																							
Sell Call Option	Strike Price (₹)	5000																							
Mr. A Receives	Premium (₹)	39																							
Buy Put Option	Strike Price (₹)	4700																							
Mr. A Pays	Premium (₹)	27																							
	Net Premium Received (₹)	12																							
	Breakeven Point (₹)	4746																							

Example:

- 1) If the price of ABC Ltd. rises to ₹ 5100 after a month, then,
 - a. Mr. A will sell the stock at ₹ 5100 earning him a profit of ₹ 342 (₹ 5100 – ₹ 4758)
 - b. Mr. A will get exercised on the Call he sold and will have to pay ₹ 100.
 - c. The Put will expire worthless.
 - d. Net premium received for the Collar is ₹ 12
 - e. Adding (a + b + d) = ₹ 342 -100 – 12 = ₹ 254.

This is the maximum return on the Collar Strategy.

However, unlike a Covered Call, the downside risk here is also limited:

- 2) If the price of ABC Ltd. falls to ₹ 4400 after a month, then,
 - a. Mr. A loses ₹ 358 on the stock ABC Ltd.
 - b. The Call expires worthless
 - c. The Put can be exercised by Mr. A and he will earn ₹ 300

d. Net premium received for the Collar is ₹ 12

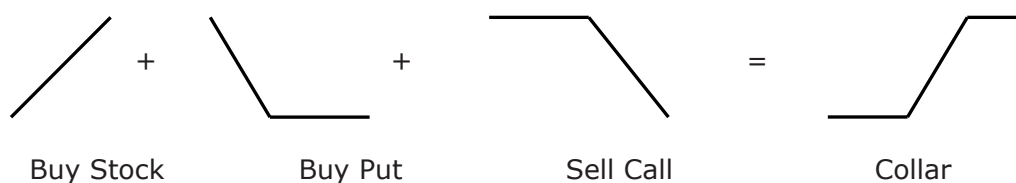
e. Adding $(a + b + d) = - ₹ 358 + 300 + 12 = - ₹ 46$

This is the maximum the investor can lose on the Collar Strategy. The upside, in this case, is much more than the downside risk.

Payoff

ABC Ltd. closes at (₹)	Payoff from Call Sold (₹)	Payoff from Put Purchased (₹)	Payoff from stock ABC Ltd.	Net payoff (₹)
4400	39	273	-358	-46
4450	39	223	-308	-46
4500	39	173	-258	-46
4600	39	73	-158	-46
4700	39	-27	-58	-46
4750	39	-27	-8	4
4800	39	-27	42	54
4850	39	-27	92	104
4858	39	-27	100	112
4900	39	-27	142	154
4948	39	-27	190	202
5000	39	-27	242	254
5050	-11	-27	292	254
5100	-61	-27	342	254
5150	-111	-27	392	254
5200	-161	-27	442	254
5248	-209	-27	490	254
5250	-211	-27	492	254
5300	-261	-27	542	254

The payoff chart (Collar)



2.11. BULL CALL SPREAD STRATEGY

BUY ITM CALL AND SELL OTM CALL

A bull call spread is constructed by buying an in-the-money (ITM) call option and selling another out-of-the-money (OTM) call option. Often the call with the lower strike price will be in-the-money while the Call with the higher strike price is out-of-the-money. Both calls must have the same underlying security and expiration month.

The net effect of the strategy is to bring down the cost and breakeven on a Buy Call (Long Call) Strategy. This strategy is exercised when an investor is moderately bullish to bullish because the investor will make a profit only when the stock price/index rises. If the stock price falls to the lower (bought) strike, the investor makes the maximum loss (cost of the trade) and if the stock price rises to the higher (sold) strike, the investor makes the maximum profit. Let us try and understand this with an example.

When to Use: Investor is moderately bullish.	Example: Mr. XYZ buys a Nifty Call with a Strike price ₹ 11000 at a premium of ₹ 170.45 and he sells a Nifty Call option with a strike price ₹ 11400 at a premium of ₹ 35.40. The net debit here is ₹ 135.05 which is also his maximum loss.		
Risk: Limited to any initial premium paid in establishing the position. Maximum loss occurs where the underlying falls to the level of the lower strike or below.	Strategy : Buy a Call with a lower strike (ITM) + Sell a Call with a higher strike (OTM)		
Reward: Limited to the difference between the two strikes minus net premium cost. Maximum profit occurs where the underlying rises to the level of the higher strike or above	Nifty index	Current Value	11100
Break-Even-Point (BEP): Strike Price of Purchased call + Net Debit Paid	Buy ITM Call Option	Strike Price (₹)	11000
	Mr. XYZ Pays	Premium (₹)	170.45
	Sell OTM Call Option	Strike Price (₹)	11400
	Mr. XYZ Receives	Premium (₹)	35.40
		Net Premium Paid (₹)	135.05
		Breakeven Point (₹)	11135.05

Payoff

On expiry Nifty Closes at	Net Payoff from Call Buy (₹)	Net Payoff from Call Sold (₹)	Net Payoff (₹)
10700	-170.45	35.4	-135.05
10750	-170.45	35.4	-135.05
10800	-170.45	35.4	-135.05
10850	-170.45	35.4	-135.05
10900	-170.45	35.4	-135.05
10950	-170.45	35.4	-135.05
11000	-170.45	35.4	-135.05
11050	-120.45	35.4	-85.05
11100	-70.45	35.4	-35.05
11135.05	-35.4	35.4	0
11200	29.55	35.4	64.95
11250	79.55	35.4	114.95
11300	129.55	35.4	164.95
11350	179.55	35.4	214.95
11400	229.55	35.4	264.95
11450	279.55	-14.6	264.95
11500	329.55	-64.6	264.95
11550	379.55	-114.6	264.95
11600	429.55	-164.6	264.95

The Bull Call Spread Strategy has

- brought the breakeven point down (if only the ₹ 11000 strike price Call was purchased the breakeven point would have been ₹ 11170.45)
- reduced the cost of the trade (if only the ₹ 11000 strike price Call was purchased the cost of the trade would have been ₹ 170.45)
- reduced the loss on the trade (if only the ₹ 11000 strike price Call was purchased the loss would have been ₹ 170.45 i.e. the premium of the Call purchased)

The payoff chart (Bull Call Spread)



2.12. BULL PUT SPREAD STRATEGY

SELL OTM PUT AND BUY FURTHER OTM PUT

A bull put spread can be profitable when the stock/index is either range bound or rising. The concept is to protect the downside of a Put sold by buying a lower strike Put, which acts as insurance for the Put sold. The lower strike Put purchased is further OTM than the higher strike Put sold ensuring that the investor receives a net credit because the Put purchased (further OTM) is cheaper than the Put sold. This strategy is equivalent to the Bull Call Spread but is done to earn a net credit (premium) and collect an income.

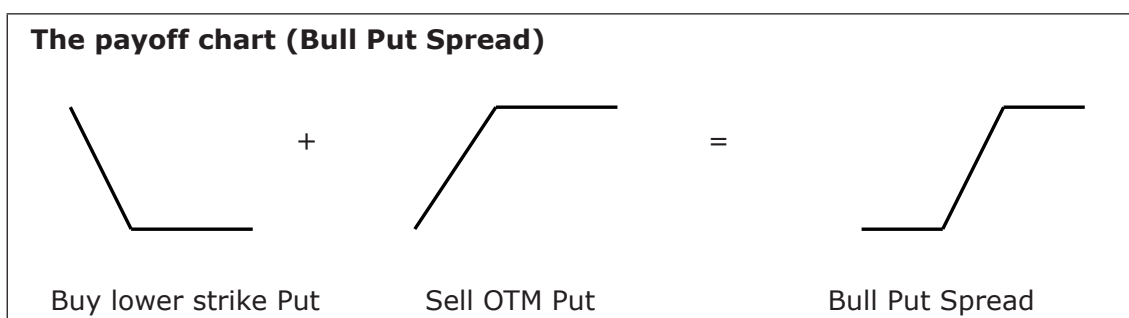
If the stock/index rises, both Puts expire worthless and the investor can retain the Premium. If the stock/index falls, then the investor's breakeven is the higher strike less the net credit received. Provided the stock remains above that level, the investor makes a profit. Otherwise, he could make a loss. The maximum loss is the difference in strikes less the net credit received. This strategy should be adopted when the stock/index trend is upward or range bound. Let us understand this with an example.

When to Use: When the investor is moderately bullish.	Example: Mr. XYZ sells a Nifty Put option with a strike price of ₹ 11000 at a premium of ₹ 21.45 and buys a further OTM Nifty Put option with a strike price ₹ 10800 at a premium of ₹ 3.00 when the current Nifty is at 11100, with both options expiring on 22nd February.		
Risk: Limited. Maximum loss occurs where the underlying falls to the level of the lower strike or below			
Reward: Limited to the net premium credit. Maximum profit occurs where underlying rises to the level of the higher strike or above.			
Breakeven: Strike Price of Short Put - Net Premium Received			
	Strategy : Sell a Put + Buy a Put		
	Nifty Index	Current Value	11100
	Sell Put Option	Strike Price (₹)	11000
	Mr. XYZ Receives	Premium (₹)	21.45
	Buy Put Option	Strike Price (₹)	10800
	Mr. XYZ Pays	Premium (₹)	3.00
		Net Premium Received (₹)	18.45
		Break Even Point (₹)	10981.55

Payoff

On expiry Nifty closes at	Net Payoff from Put bought (₹)	Net Payoff from Put sold (₹)	Net Payoff (₹)
10400	397	-578.55	-181.55
10500	297	-478.55	-181.55
10600	197	-378.55	-181.55
10700	97	-278.55	-181.55
10800	-3	-178.55	-181.55
10900	-3	-78.55	-81.55
10981.55	-3	3	0
11100	-3	21.45	18.45
11200	-3	21.45	18.45
11300	-3	21.45	18.45
11400	-3	21.45	18.45
11500	-3	21.45	18.45
11600	-3	21.45	18.45
11700	-3	21.45	18.45
11800	-3	21.45	18.45

The strategy earns a net income for the investor as well as limits the downside risk of a Put sold.



2.13. BEAR CALL SPREAD STRATEGY SELL ITM CALL AND BUY OTM CALL

The Bear Call Spread strategy can be adopted when the investor feels that the stock/index is either range bound or falling. The concept is to protect the downside of a Call Sold by buying a Call of a higher strike price to insure the Call sold. In this strategy, the investor receives a net credit because the Call he buys is of a higher strike price than the Call sold. The strategy requires the investor to buy out-of-the-money (OTM) call options while simultaneously selling in-the-money (ITM) call options on the same underlying stock index. This strategy can also be done with both OTM calls with the Call purchased being higher OTM strike than the Call

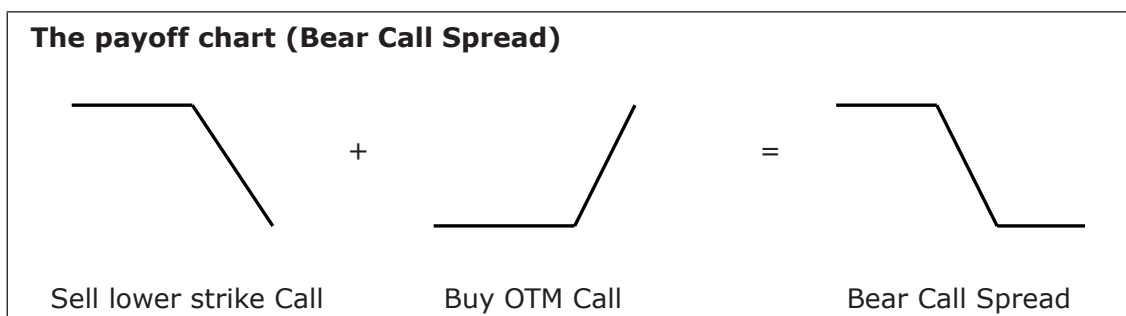
sold. If the stock/index falls both Calls will expire worthless and the investor can retain the net credit. If the stock/index rises then the breakeven is the lower strike plus the net credit. Provided the stock remains below that level, the investor makes a profit. Otherwise, he could make a loss. The maximum loss is the difference in strikes less the net credit received. Let us understand this with an example.

When to use: When the investor is mildly bearish on market. Risk: Limited to the difference between the two strikes minus the net premium. Reward: Limited to the net premium received for the position i.e., the premium received for the short call minus the premium paid for the long call. Breakeven Point: Lower Strike + Net credit	Example: Mr. XYZ is bearish on ABC Ltd. He sells an ITM call option with strike price of ₹ 2600 at a premium of ₹ 154 and buys an OTM call option with strike price ₹ 2800 at a premium of ₹ 49. <table><tr><th colspan="3">Strategy : Sell a Call with a lower strike (ITM) + Buy a Call with a higher strike (OTM)</th></tr><tr><td>ABC Ltd.</td><td>Current Value</td><td>2694</td></tr><tr><td>Sell ITM Call Option</td><td>Strike Price (₹)</td><td>2600</td></tr><tr><td>Mr. XYZ receives</td><td>Premium (₹)</td><td>154</td></tr><tr><td>Buy OTM Call Option</td><td>Strike Price (₹)</td><td>2800</td></tr><tr><td>Mr. XYZ pays</td><td>Premium (₹)</td><td>49</td></tr><tr><td></td><td>Net premium received (₹)</td><td>105</td></tr><tr><td></td><td>Breakeven Point (₹)</td><td>2705</td></tr></table>	Strategy : Sell a Call with a lower strike (ITM) + Buy a Call with a higher strike (OTM)			ABC Ltd.	Current Value	2694	Sell ITM Call Option	Strike Price (₹)	2600	Mr. XYZ receives	Premium (₹)	154	Buy OTM Call Option	Strike Price (₹)	2800	Mr. XYZ pays	Premium (₹)	49		Net premium received (₹)	105		Breakeven Point (₹)	2705
Strategy : Sell a Call with a lower strike (ITM) + Buy a Call with a higher strike (OTM)																									
ABC Ltd.	Current Value	2694																							
Sell ITM Call Option	Strike Price (₹)	2600																							
Mr. XYZ receives	Premium (₹)	154																							
Buy OTM Call Option	Strike Price (₹)	2800																							
Mr. XYZ pays	Premium (₹)	49																							
	Net premium received (₹)	105																							
	Breakeven Point (₹)	2705																							

Payoff

On expiry ABC Ltd. Closes at	Net Payoff from Call Sold (₹)	Net Payoff from Call bought (₹)	Net Payoff (₹)
2100	154	-49	105
2200	154	-49	105
2300	154	-49	105
2400	154	-49	105
2500	154	-49	105
2600	154	-49	105
2700	54	-49	5
2705	49	-49	0
2800	-46	-49	-95
2900	-146	51	-95
3000	-246	151	-95
3100	-346	251	-95
3200	-446	351	-95
3300	-546	451	-95

The strategy earns a net income for the investor as well as limits the downside risk of a Call sold.



2.14. BEAR PUT SPREAD BUY ITM PUT AND SELL OTM PUT

This strategy requires the investor to buy an in-the-money (higher) put option and sell an out-of-the-money (lower) put option on the same stock with the same expiration date. This strategy creates a net debit for the investor. The net effect of the strategy is to bring down the cost and raise the breakeven on buying a Put (Long Put). The strategy needs a Bearish outlook since the investor will make money only when the stock price/index falls. The bought Puts will have the effect of capping the investor's downside. While the Puts sold will reduce the investors costs, risk and raise the breakeven point (from Put exercise point of view). If the stock price closes below the out-of-the-money (lower) put option strike price on the expiration date, then the investor reaches maximum profits. If the stock price increases above the in-the-money (higher) put option strike price at the expiration date, then the investor has a maximum loss potential of the net debit.

When to use: When you are moderately bearish on market direction	Example: ABC Ltd. is presently at 2694. Mr. XYZ expects ABC Ltd. share price to fall. He buys one ABC Ltd. ITM Put with a strike price ₹ 2800 at a premium of ₹ 132 and sells one ABC Ltd. OTM Put with strike price ₹ 2600 at a premium ₹ 52.	
Risk: Limited to the net amount paid for the spread i.e. the premium paid for long position less premium received for short position.	Strategy : BUY A PUT with a higher strike (ITM) + SELL A PUT with a lower strike (OTM)	
Reward: Limited to the difference between the two strike prices minus the net premium paid for the position.	ABC Ltd.	Current Value 2694
Break Even Point: Strike Price of Long Put - Net Premium Paid	Buy ITM Put Option	Strike Price (₹) 2800
	Mr. XYZ pays	Premium (₹) 132
	Sell OTM Put Option	Strike Price (₹) 2600
	Mr. XYZ receives	Premium (₹) 52
		Net Premium Paid (₹) 80
		Breakeven Point (₹) 2720

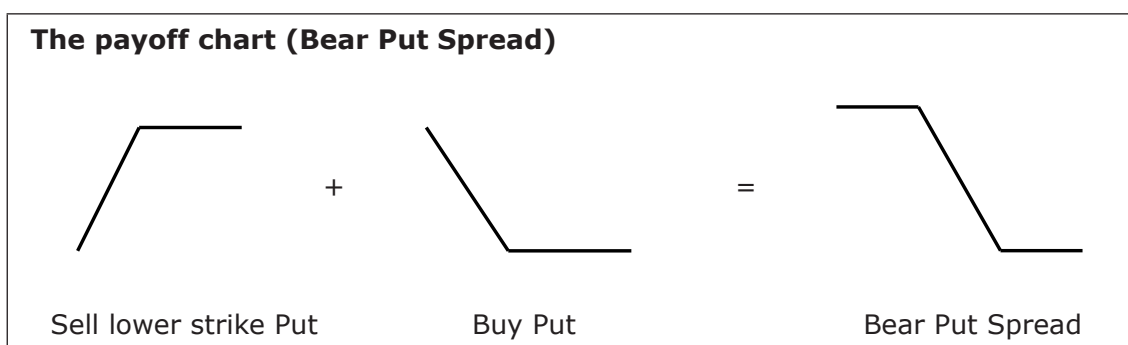
Payoff

On expiry, ABC Ltd. closes at	Net Payoff from Put Buy (₹)	Net Payoff from Put Sold (₹)	Net payoff (₹)
2200	468	-348	120
2300	368	-248	120
2400	268	-148	120
2500	168	-48	120
2600	68	52	120
2720	-52	52	0
2700	-32	52	20
2800	-132	52	-80
2900	-132	52	-80
3000	-132	52	-80
3100	-132	52	-80

The Bear Put Spread Strategy has

- raised the breakeven point (if only the ₹ 2800 strike price Put was purchased the breakeven point would have been ₹ 2668),
- reduced the cost of the trade (if only the ₹ 2800 strike price Put was purchased the cost of the trade would have been ₹ 132),
- reduced the loss on the trade (if only the ₹ 2800 strike price Put was purchased the loss would have been ₹ 132 i.e. the premium of the Put purchased).

However, the strategy also has limited gains and is therefore ideal when markets are moderately bearish.



2.15. LONG CALL BUTTERFLY SELL 2 ATM CALLS, BUY 1 ITM CALL AND BUY 1 OTM CALL

A Long Call Butterfly is to be adopted when the investor is expecting very little movement in the stock price/index. The investor is looking to gain from low volatility at a low cost. The strategy offers a good risk/reward ratio, together with low cost. A long butterfly is similar

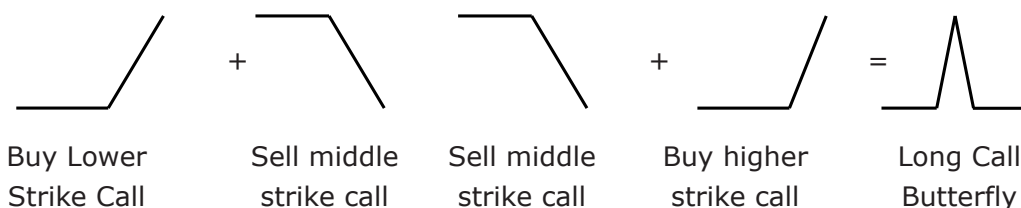
to a Short Straddle except your losses are limited. The strategy can be done by selling 2 ATM Calls, buying 1 ITM Call, and buying 1 OTM Call options (there should be equidistance between the strike prices). The result is positive in case the stock/index remains range bound. The maximum reward in this strategy is however restricted and takes place when the stock/index is at the middle strike at expiration. The maximum losses are also limited. Let us see an example to understand the strategy.

When to use: When the investor is neutral on market direction and bearish on volatility .	Example:		
Risk Net debit paid.	Nifty is at 11000. Mr. XYZ expects very little movement in the share price. He sells 2 ATM Nifty Call Options with a strike price of ₹ 11000 at a premium of ₹ 97.90 each, buys 1 ITM Nifty Call Option with a strike price of ₹ 10900 at a premium of ₹ 141.55 and buys 1 OTM Nifty Call Option with a strike price of ₹ 11100 at a premium of ₹ 64. The Net debit is ₹ 9.75.		
Reward Difference between adjacent strikes minus net debit			
Break Even Point:	STRATEGY : SELL 2 ATM CALL, BUY 1 ITM CALL OPTION AND BUY 1 OTM CALL OPTION		
Upper Breakeven Point =	Nifty index	Current Value	11000
Strike Price of Higher	Sell 2 ATM Call Option	Strike Price (₹)	11000
Strike Long Call-Net	Mr. XYZ receives	Premium (₹)	195.80
Premium Paid	Buy 1 ITM Call Option	Strike Price (₹)	10900
	Mr. XYZ pays	Premium (₹)	141.55
Lower Breakeven Point =	Buy 1 OTM Call Option	Strike Price (₹)	11100
Strike Price of Lower Strike	Mr. XYZ pays	Premium (₹)	64
Long Call + Net Premium		Breakeven Point (Upper) (₹)	11090.25
Paid		Breakeven Point (Lower) (₹)	10909.75

Payoff

On expiry Nifty Closes at	Net Payoff from 2 ATM Calls Sold (₹)	Net Payoff from 1 ITM Call purchased (₹)	Net Payoff from 1 OTM Call purchased (₹)	Net Payoff (₹)
10700	195.8	-141.55	-64	-9.75
10750	195.8	-141.55	-64	-9.75
10800	195.8	-141.55	-64	-9.75
10850	195.8	-141.55	-64	-9.75
10909.75	195.8	-131.8	-64	0
10950	195.8	-91.55	-64	40.25
11000	195.8	-41.55	-64	90.25
11050	95.8	8.45	-64	40.25
11090.25	15.3	48.7	-64	0
11150	-104.2	108.45	-14	-9.75
11200	-204.2	158.45	36	-9.75
11250	-304.2	208.45	86	-9.75
11300	-404.2	258.45	136	-9.75

The payoff chart (Long Call Butterfly)



2.16. SHORT CALL BUTTERFLY

**BUY 2 ATM CALLS,
SELL 1 ITM CALL AND
SELL 1 OTM CALL**

A Short Call Butterfly is a strategy for volatile markets. It is the opposite of Long Call Butterfly, which is a range bound strategy. The Short Call Butterfly can be constructed by Selling one lower striking in-the-money Call, buying two at-the-money Calls and selling another higher strike out-of-the-money Call, giving the investor a net credit (therefore it is an income strategy). There should be equal distance between each strike. The resulting position will be profitable in case there is a big move in the stock/index. The maximum risk occurs if the stock/index is at the middle strike at expiration. The maximum profit occurs if the stock finishes on either side of the upper and lower strike prices at expiration. However, this

strategy offers very small returns when compared to straddles, strangles with only slightly less risk. Let us understand this with an example.

When to use: You are neutral on market direction and bullish on volatility. Neutral means that you expect the market to move in either direction - i.e. bullish and bearish. Risk Limited to the net difference between the adjacent strikes (₹100 in this example) less the premium received for the position. Reward Limited to the net premium received for the option spread. Break Even Point: Upper Breakeven Point = Strike Price of Highest Strike Short Call- Net Premium Received Lower Breakeven Point = Strike Price of Lowest Strike Short Call + Net Premium Received	Example: Nifty is at 11000. Mr. XYZ expects large volatility in the Nifty irrespective of which direction the movement is, upwards or downwards. Mr. XYZ buys 2 ATM Nifty Call Options with a strike price of ₹ 11000 at a premium of ₹ 97.90 each, sells 1 ITM Nifty Call Option with a strike price of ₹ 10900 at a premium of ₹ 141.55 and sells 1 OTM Nifty Call Option with a strike price of ₹ 11100 at a premium of ₹ 64. The Net Credit is ₹ 9.75. <table><tr><th colspan="3">STRATEGY</th></tr><tr><th colspan="3">BUY 2 ATM CALL OPTIONS, SELL 1 ITM CALL OPTION AND SELL 1 OTM CALL OPTION.</th></tr><tr><td>Nifty index</td><td>Current Market Price</td><td>11000</td></tr><tr><td>Buy 2 ATM Call Option</td><td>Strike Price (₹)</td><td>11000</td></tr><tr><td>Mr. XYZ pays</td><td>Premium (₹)</td><td>195.80</td></tr><tr><td>Sells 1 ITM Call Option</td><td>Strike Price (₹)</td><td>10900</td></tr><tr><td>Mr. XYZ receives</td><td>Premium (₹)</td><td>141.55</td></tr><tr><td>Sells 1 OTM Call Option</td><td>Strike Price (₹)</td><td>11100</td></tr><tr><td>Mr. XYZ receives</td><td>Premium (₹)</td><td>64</td></tr><tr><td></td><td>Break Even Point (Upper) (₹)</td><td>10909.75</td></tr><tr><td></td><td>Break Even Point (Lower) (₹)</td><td>11090.25</td></tr></table>	STRATEGY			BUY 2 ATM CALL OPTIONS, SELL 1 ITM CALL OPTION AND SELL 1 OTM CALL OPTION.			Nifty index	Current Market Price	11000	Buy 2 ATM Call Option	Strike Price (₹)	11000	Mr. XYZ pays	Premium (₹)	195.80	Sells 1 ITM Call Option	Strike Price (₹)	10900	Mr. XYZ receives	Premium (₹)	141.55	Sells 1 OTM Call Option	Strike Price (₹)	11100	Mr. XYZ receives	Premium (₹)	64		Break Even Point (Upper) (₹)	10909.75		Break Even Point (Lower) (₹)	11090.25
STRATEGY																																		
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Sells 1 OTM Call Option	Strike Price (₹)	11100																																
Mr. XYZ receives	Premium (₹)	64																																
	Break Even Point (Upper) (₹)	10909.75																																
	Break Even Point (Lower) (₹)	11090.25																																

Payoff

On expiry Nifty Closes at	Net Payoff from 2 ATM Calls Bought (₹)	Net Payoff from 1 ITM Call Sold (₹)	Net Payoff from 1 OTM Call Sold (₹)	Net Payoff (₹)
10700	-195.8	141.55	64	9.75
10750	-195.8	141.55	64	9.75
10800	-195.8	141.55	64	9.75
10850	-195.8	141.55	64	9.75
10909.75	-195.8	131.8	64	0
10950	-195.8	91.55	64	-40.25
11000	-195.8	41.55	64	-90.25
11050	-95.8	-8.45	64	-40.25
11090.25	-15.3	-48.7	64	0
11150	104.2	-108.45	14	9.75
11200	204.2	-158.45	-36	9.75
11250	304.2	-208.45	-86	9.75
11300	404.2	-258.45	-136	9.75
11350	504.2	-308.45	-186	9.75
11400	604.2	-358.45	-236	9.75

The payoff chart (Short Call Butterfly)



2.17. LONG CALL CONDOR

**BUY 1 ITM CALL (LOWER STRIKE),
SELL 1 ITM CALL (LOWER MIDDLE),
SELL 1 OTM CALL (HIGHER MIDDLE),
BUY 1 OTM CALL (HIGHER STRIKE)**

A Long Call Condor is very similar to a long butterfly strategy. The difference is that the two middle sold options have different strikes. The profitable area of the payoff profile is wider than that of the Long Butterfly (see pay- off diagram).

The strategy is suitable in a range bound market. The Long Call Condor involves buying 1 ITM Call (lower strike), selling 1 ITM Call (lower middle), selling 1 OTM call (higher middle)

and buying 1 OTM Call (higher strike). The long options at the outside strikes ensure that the risk is capped on both the sides. The resulting position is profitable if the stock/index remains range bound and shows very little volatility. The maximum profits occur if the stock finishes between the middle strike prices at expiration. Let us understand this with an example.

When to Use: When an investor believes that the underlying market will trade in a range with low volatility until the options expire. Risk Limited to the minimum of the difference between the lower strike call spread less the higher call spread less the total premium paid for the condor. Reward Limited. The maximum profit of a long condor will be realized when the stock is trading between the two middle strike prices. Break Even Point: Upper Breakeven Point = Highest Strike – Net Debit Lower Breakeven Point = Lowest Strike + Net Debit	Example: Nifty is at 11000. Mr. XYZ expects little volatility in the Nifty and expects the market to remain range bound. Mr. XYZ buys 1 ITM Nifty Call Options with a strike price of ₹ 10800 at a premium of ₹ 241, sells 1 ITM Nifty Call Option with a strike price of ₹ 10900 at a premium of ₹ 126, sells 1 OTM Nifty Call Option with a strike price of ₹ 11100 at a premium of ₹ 83 and buys 1 OTM Nifty Call Option with a strike price of ₹ 11200 at a premium of ₹ 51. The Net debit is ₹ 83 which is also the maximum possible loss. <table><tr><td colspan="3">STRATEGY : BUY 1 ITM CALL OPTION (LOWER STRIKE), SELL 1 ITM CALL OPTION (LOWER MIDDLE), SELL 1 OTM CALL OPTION (HIGHER MIDDLE), BUY 1 OTM CALL OPTION (HIGHER STRIKE)</td></tr><tr><td>Nifty index</td><td>Current Value</td><td>11000</td></tr><tr><td>Buy 1 ITM Call Option</td><td>Strike Price (₹)</td><td>10800</td></tr><tr><td>Mr. XYZ pays</td><td>Premium (₹)</td><td>241</td></tr><tr><td>Sell 1 ITM Call Option</td><td>Strike Price (₹)</td><td>10900</td></tr><tr><td>Mr. XYZ receives</td><td>Premium (₹)</td><td>126</td></tr><tr><td>Sell 1 OTM Call Option</td><td>Strike Price (₹)</td><td>11100</td></tr><tr><td>Mr. XYZ receives</td><td>Premium (₹)</td><td>83</td></tr><tr><td>Buy 1 OTM Call Option</td><td>Strike Price (₹)</td><td>11200</td></tr><tr><td>Mr. XYZ pays</td><td>Premium (₹)</td><td>51</td></tr><tr><td></td><td>Break Even Point (Upper) (₹)</td><td>11117</td></tr><tr><td></td><td>Break Even Point (Lower) (₹)</td><td>10883</td></tr></table>	STRATEGY : BUY 1 ITM CALL OPTION (LOWER STRIKE), SELL 1 ITM CALL OPTION (LOWER MIDDLE), SELL 1 OTM CALL OPTION (HIGHER MIDDLE), BUY 1 OTM CALL OPTION (HIGHER STRIKE)			Nifty index	Current Value	11000	Buy 1 ITM Call Option	Strike Price (₹)	10800	Mr. XYZ pays	Premium (₹)	241	Sell 1 ITM Call Option	Strike Price (₹)	10900	Mr. XYZ receives	Premium (₹)	126	Sell 1 OTM Call Option	Strike Price (₹)	11100	Mr. XYZ receives	Premium (₹)	83	Buy 1 OTM Call Option	Strike Price (₹)	11200	Mr. XYZ pays	Premium (₹)	51		Break Even Point (Upper) (₹)	11117		Break Even Point (Lower) (₹)	10883
STRATEGY : BUY 1 ITM CALL OPTION (LOWER STRIKE), SELL 1 ITM CALL OPTION (LOWER MIDDLE), SELL 1 OTM CALL OPTION (HIGHER MIDDLE), BUY 1 OTM CALL OPTION (HIGHER STRIKE)																																					
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Payoff

On expiry, Nifty Closes at	Net Payoff from 1 ITM Call purchased (₹)	Net Payoff from 1 ITM Call sold (₹)	Net Payoff from 1 OTM Call sold (₹)	Net Payoff from 1 OTM Call purchased (₹)	Net Payoff (₹)
10700	-241	126	83	-51	-83
10750	-241	126	83	-51	-83
10800	-241	126	83	-51	-83
10850	-191	126	83	-51	-33
10883	-158	126	83	-51	0
10950	-91	76	83	-51	17
11000	-41	26	83	-51	17
11050	9	-24	83	-51	17
11117	76	-91	66	-51	0
11150	109	-124	33	-51	-33
11200	159	-174	-17	-51	-83
11250	209	-224	-67	-1	-83
11300	259	-274	-117	49	-83

Example:

Suppose Nifty is at 11000 in June. An investor enters a condor trade by buying a ₹ 10800 strike price call at a premium of ₹ 241, sells a ₹ 10900 strike price call at a premium of ₹ 126, sells another call at a strike price of ₹ 11100 at a premium of ₹ 83 and buys a call at a strike price of ₹ 11200 at a premium of ₹ 51. The net debit from the trades is ₹ 83. This is also his maximum loss.

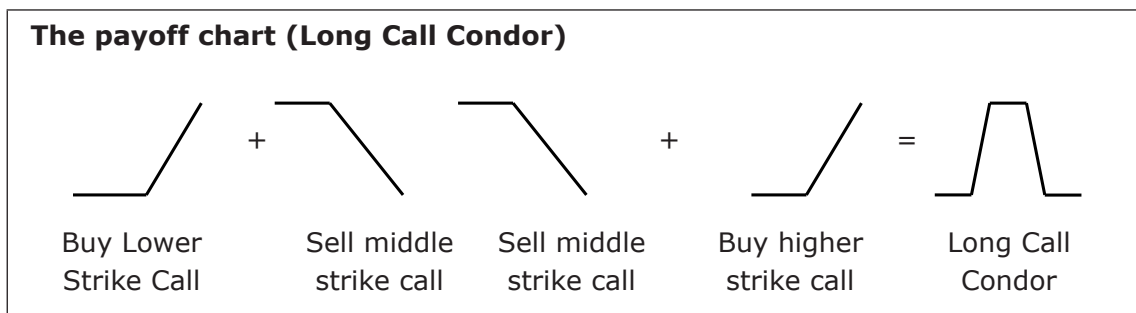
To further see why ₹ 83 is his maximum possible loss, let's examine what happens when Nifty falls to 10700 or rises to 11300 on expiration.

At 10700, all the options expire worthless, so the initial debit taken of ₹ 83 is the investor's maximum loss.

At 11300, the long ₹ 10800 call earns ₹ 259 (₹ 11300 – ₹ 10800 – ₹ 241). The two calls sold result in a loss of ₹ 391 (The call with a strike price of ₹ 10900 makes a loss of ₹ 274 and the call with a strike price of ₹ 11100 makes a loss of ₹ 117). Finally, the call purchased with a strike price of ₹ 11200 earns ₹ 49 (₹ 11300 – ₹ 11200 – ₹ 51). Total loss (₹ 259 – ₹ 391 + ₹ 49) works out to ₹ 83. Thus, the long condor trader still suffers the maximum loss that is equal to the initial debit taken when entering the trade.

If during the time of expiry of the contract, Nifty is still at 11000, the ₹ 10900 strike price call and ₹ 11100 strike price call sold earns money while the ₹ 10800 strike price call and ₹ 11200 strike price call purchased end in losses. Such that the total gain comes to ₹ 17 which is also the maximum gain the investor can make with this strategy.

The maximum profit for the condor trade may be low in relation to other trading strategies but it has a comparatively wider profit zone. In this example, profit is achieved if the Nifty at expiry is anywhere between ₹ 10883 and ₹ 11117.



2.18. SHORT CALL CONDOR

**SHORT 1 ITM CALL (LOWER STRIKE),
LONG 1 ITM CALL (LOWER MIDDLE),
LONG 1 OTM CALL (HIGHER MIDDLE),
SHORT 1 OTM CALL (HIGHER STRIKE)**

A Short Call Condor is very similar to a short butterfly strategy. The difference is that the two middle bought options have different strikes. The strategy is suitable for a volatile market. The Short Call Condor involves selling 1 ITM Call (lower strike), buying 1 ITM Call (lower middle), buying 1 OTM call (higher middle) and selling 1 OTM Call (higher strike). The resulting position is profitable if the stock/index shows very high volatility and there is a big move in the stock k / index. The maximum profits occur if the stock/index finishes on either side of the upper or lower strike prices at expiration. Let us understand this with an example.

When to Use: When an investor believes that the underlying market will break out of a trading range but is not sure in which direction. Risk Limited. The maximum loss of a short condor occurs at the centre of the option spread. Reward Limited. The maximum profit of a short condor occurs when the underlying stock/index is trading past the upper or lower strike prices. Break Even Point: Upper Breakeven Point = Highest Strike – Net Credit Lower Breakeven Point = Lowest Strike + Net Credit	Example: Nifty is at 11000. Mr. XYZ expects high volatility in the Nifty and expects the market to break open significantly on any side. Mr. XYZ sells 1 ITM Nifty Call Options with a strike price of ₹ 10800 at a premium of ₹ 241, buys 1 ITM Nifty Call Option with a strike price of ₹ 10900 at a premium of ₹ 126, buys 1 OTM Nifty Call Option with a strike price of ₹ 11100 at a premium of ₹ 83 and sells 1 OTM Nifty Call Option with a strike price of ₹ 11200 at a premium of ₹ 51. The Net credit is of ₹ 83. <table><tr><td colspan="3">STRATEGY : SHORT 1 ITM CALL OPTION (LOWER STRIKE), LONG 1 ITM CALL OPTION (LOWER MIDDLE), LONG 1 OTM CALL OPTION (HIGHER MIDDLE), SHORT 1 OTM CALL OPTION (HIGHER STRIKE)</td></tr><tr><td>Nifty index</td><td>Current Value</td><td>11000</td></tr><tr><td>Sell 1 ITM Call Option</td><td>Strike Price (₹)</td><td>10800</td></tr><tr><td></td><td>Premium (₹)</td><td>241</td></tr><tr><td>Buy 1 ITM Call Option</td><td>Strike Price (₹)</td><td>10900</td></tr><tr><td></td><td>Premium (₹)</td><td>126</td></tr><tr><td>Buy 1 OTM Call Option</td><td>Strike Price (₹)</td><td>11100</td></tr><tr><td></td><td>Premium (₹)</td><td>83</td></tr><tr><td>Sell 1 OTM Call Option</td><td>Strike Price (₹)</td><td>11200</td></tr><tr><td></td><td>Premium (₹)</td><td>51</td></tr><tr><td></td><td>Breakeven Point (Upper) (₹)</td><td>11117</td></tr><tr><td></td><td>Breakeven Point (Lower) (₹)</td><td>10883</td></tr></table>	STRATEGY : SHORT 1 ITM CALL OPTION (LOWER STRIKE), LONG 1 ITM CALL OPTION (LOWER MIDDLE), LONG 1 OTM CALL OPTION (HIGHER MIDDLE), SHORT 1 OTM CALL OPTION (HIGHER STRIKE)			Nifty index	Current Value	11000	Sell 1 ITM Call Option	Strike Price (₹)	10800		Premium (₹)	241	Buy 1 ITM Call Option	Strike Price (₹)	10900		Premium (₹)	126	Buy 1 OTM Call Option	Strike Price (₹)	11100		Premium (₹)	83	Sell 1 OTM Call Option	Strike Price (₹)	11200		Premium (₹)	51		Breakeven Point (Upper) (₹)	11117		Breakeven Point (Lower) (₹)	10883
STRATEGY : SHORT 1 ITM CALL OPTION (LOWER STRIKE), LONG 1 ITM CALL OPTION (LOWER MIDDLE), LONG 1 OTM CALL OPTION (HIGHER MIDDLE), SHORT 1 OTM CALL OPTION (HIGHER STRIKE)																																					
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Payoff

On expiry Nifty Closes at	Net Payoff from 1 ITM Call sold (₹)	Net Payoff from 1 ITM Call purchased (₹)	Net Payoff from 1 OTM Call purchased (₹)	Net Payoff from 1 OTM Call sold (₹)	Net Payoff (₹)
10700	241	-126	-83	51	83
10750	241	-126	-83	51	83
10800	241	-126	-83	51	83
10850	191	-126	-83	51	33
10883	158	-126	-83	51	0
10950	91	-76	-83	51	-17
11000	41	-26	-83	51	-17
11050	-9	24	-83	51	-17
11117	-76	91	-66	51	0
11150	-109	124	-33	51	33
11200	-159	174	17	51	83
11250	-209	224	67	1	83
11300	-259	274	117	-49	83
11350	-309	324	167	-99	83
11400	-359	374	217	-149	83

The payoff chart (Short Call Condor)



3. VOLATILITY

Volatility is the degree of variation of a price time series, such as the closing prices of a particular stock. This is measured by the standard deviation of the time series, a statistical measure of dispersement. Volatility is not concerned with the direction of change, but with the amount of change.

3.1. TYPES OF VOLATILITY

Now that we have understood the essence of volatility, let us look at its different types.

Historical Volatility

It gauges the fluctuations of underlying securities by measuring the price changes over a predetermined period of time in the past. The calculation can also be done on an intra-day basis, but this process is generally done over the closing prices of securities. Based on the intended duration of the options trade, one can calculate the historical volatility.

Implied Volatility

Another type of volatility that is extensively used in options trading is Implied Volatility. In the context of options, the implied volatility indicates the extent to which the return of the underlying asset may fluctuate between now and the expiration date of the option. The higher the implied volatility the more people think the price of the underlying asset will move a lot. Implied volatility is a result of people buying and selling in the market, and is a forward-looking imperfect metric, whereas historical volatility looks at a particular time frame in the past and measures the movement in that duration.

Realized Volatility

Unlike implied volatility which tells us the expected variance in a future time period, realized volatility tells us the actual volatility realized for that future period of time. Naturally, you can calculate the realized volatility only after that future time period. Generally, your outlook on the difference between implied volatility, what people are expecting, and realized volatility, what actually will happen, will help you to select the right options trading strategies.

3.2. CALCULATING THE HISTORICAL VOLATILITY

1. Compute the log returns for the time series

$$R = \ln\left(\frac{C}{C_{t-1}}\right)$$

No.	Date	Nifty Close Price (C)	Log returns (R)
1	19-Jan-18	10921	
2	22-Jan-18	10992.5	0.0065
3	23-Jan-18	11107.85	0.0104
4	24-Jan-18	11105.4	-0.0002
5	25-Jan-18	11062.6	-0.0039
6	29-Jan-18	11137.65	0.0068
7	30-Jan-18	11072.1	-0.0059
8	31-Jan-18	11055.4	-0.0015
9	01-Feb-18	11031.3	-0.0022
10	02-Feb-18	10755.85	-0.0253

2. Compute the standard deviation of the log returns

- A. Find the mean

$$u = \frac{\sum \ln\left(\frac{C}{C_{t-1}}\right)}{N} = \frac{-0.0152}{9} = -0.0017$$

- B. For each data point, find the square of its distance to the mean and sum it

$$\sum (R - u)^2 = 0.00086774$$

- C. Divide by the number of data points - 1

$$\frac{\sum (R - u)^2}{N - 1} = \frac{0.00086774}{8} = 0.000108$$

- D. Take the square root

$$\sqrt{\frac{\sum (R - u)^2}{N - 1}} \times 100 = 1.04\%$$

- E. Calculate annualized volatility

$$\sqrt{\frac{\sum (R - u)^2}{N - 1}} \times 100 \times \sqrt{252} = 16.5\%$$

Thus, 16.5% is the annualized volatility computed using 10 days of data. With more data points we will get better estimate of historical volatility.

3.3. IMPACT OF VOLATILITY ON OPTIONS

Often while entering into options trading strategies, the volatility is often ignored by most of them. The relationship of Volatility and the options strategies learned earlier is therefore very important. More of this concept will be discussed in the Options Trading (Advanced) Module but for now, as the volatility increases, everything else remaining the same, the price of the option increases. Similarly, if the volatility drops then the price of the option also decreases. That is to say, if you had bought a Nifty 11000 strike call option and implied volatility declines then, everything else being the same, the price of the call option will decline. The below table summarizes the impact of the implied volatility on the options.

Option Type	Rise in Implied Volatility	Fall in Implied Volatility
Long Call	Positive	Negative
Short Call	Negative	Positive
Long Put	Positive	Negative
Short Put	Negative	Positive

Summary

In this workbook, we have learned basics of options, various options strategies, and different types of volatility. There are two types of options: calls and puts. A call option gives the buyer the right to buy an asset by a certain date for a certain price. A put option gives the buyer the right to sell an asset by a certain date for a certain price. There are various options trading strategies to take advantage of bullish, bearish, or neutral view on the underlying asset. The implied volatility of a stock is a measure of how uncertain we are about the expected future stock price movements. As the implied volatility increases, the chance that the stock can move a lot in either direction increases and the price of both calls and puts increases given everything else being the same. Therefore, the view about the expected volatility is very useful to select the correct option trading strategies.

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